Statistics Department
Arabic Section
Fourth Level

Predictive Modeling of Employee Compensation: A Statistical Analysis of HR Data Using Regression and Machine Learning Techniques

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Table Of Content:

Introduction:

1.Background and Purpose

In the modern corporate landscape, Human Resources (HR) analytics has emerged as a vital tool for making objective, data-driven decisions regarding a company's most asset: its workforce. Organizations today systematically analyze employee data to uncover hidden patterns in compensation and career progression, which in turn inform strategic talent management initiatives and workforce equity.

2. Research Scope and Variable Selection

This study utilizes a comprehensive dataset to examine how specific individual attributes influence financial outcomes. While many variables exist within HR data, this analysis focuses on eight strategically selected dimensions to address fundamental questions about salary growth and organizational structure:

- **Identifier:** emp.no
- **Demographics:** Age, Gender
- **Professional Background:** Total Working Years, Education, Department
- **Status & Growth:** Monthly Income, Years Since Last Promotion

3. Analytical Objectives

By examining these core dimensions, the analysis provides actionable insights into several critical HR domains:

- **Compensation Analysis:** Evaluating how income is distributed across demographic groups and experience levels.
- **Career Progression Patterns:** Identifying trends in promotional velocity and potential career stagnation.
- **Organizational Composition:** Analyzing the allocation of human capital across different functional areas (Departments).
- **Workforce Demographics:** Understanding the age, gender, and educational profile of the organization.

4. Research Methodology and Primary Question

By applying advanced statistical modeling specifically **Multiple Linear Regression** and **Machine Learning (Decision Trees)**, this research moves beyond simple descriptions to provide predictive insights into the mechanics of employee pay.

Central to this investigation is the following primary research question:

"To what extent do demographic, educational, and professional factors collectively predict an employee's monthly income, and which of these factors serve as the most influential driver of compensation?"

Data Description:

Source of the data:

The data used for this analysis was obtained from the "HR Analytics Dataset" published on Kaggle by the user anshika2301. This dataset is a collection of employee-related records used primarily for analyzing employee attrition, satisfaction, and performance metrics. The dataset is licensed under CC0: Public Domain, meaning it is free to use for analysis and reporting.

Number of observations:

The dataset consists of **1,470** observations, where each observation represents an individual employee's record.

Target population:

The target population for this study consists of corporate employees from a mid-to-large sized organization. The sample includes 1,480 individuals with diverse backgrounds in terms of age, gender, and education, distributed across three primary departments: Sales, Research & Development, and Human Resources. This population represents a typical corporate workforce used to analyze factors influencing employee turnover and job satisfaction.

The Variables Measured:

Variable Name	Description	Type of Data
Emp.no	A unique identification number assigned to each employee.	**Identifier** (Nominal)
Age	The chronological age of the employee in years.	**Numeric** (Continuous)
Gender	The gender identity of the employee (Male, Female).	**Categorical** (Binary)
Education	The level of education attained by the employee, coded 1–5 (1: high school, 2: Associate degree, 3: Bachelor’s Degree, 4: Master's Degree, 5: Doctoral Degree).	**Ordinal** (Ranked)

Variable Name	Description	Type of Data
Department	The specific division where the employee works (Sales, R&D, HR).	**Categorical** (Nominal)
Total Working Years	Total number of years the employee has been in the workforce (experience).	**Numeric** (Discrete)
Years Since Last Promotion	Time elapsed since the employee was last promoted within the company.	**Numeric** (Discrete)
Monthly Income	The gross monthly salary of the employee.	**Numeric** (Ratio)

- **Employee Identity (Emp.no):** Used as a unique key to ensure no duplicate records are analyzed.
- **Demographic Profile (Age, Gender, Education):** These variables allow for an analysis of the workforce composition and help identify if age or education level correlates with career progression.
- **Professional Experience (Total Working Years, Years Since Last Promotion):** These metrics represent the employee's career maturity and the "stagnation" factor. For example, a high number of years since a promotion may indicate a risk of turnover.
- **Organizational Structure (Department):** This helps compare whether certain behaviors (like salary growth or promotion speed) differ between technical and administrative roles.
- **Financial Compensation (Monthly Income):** This serves as a primary indicator of employee value and is often the most significant factor in attrition and satisfaction studies.

<u>Data Preview:</u>

This "head" of the data illustrates how employee records are structured across the eight selected variables.

```
   emp.no Age Monthly.Income Gender Department Total.Working.Years Years.Since.Last.Promotion         Education
1 STAFF-1  41           5993 Female      Sales                   8                         0 Associates Degree
2 STAFF-2  49           5130   Male        R&D                  10                         1       High School
3 STAFF-4  37           2090   Male        R&D                   7                         0 Associates Degree
4 STAFF-5  33           2909 Female        R&D                   8                         3   Master's Degree
5 STAFF-7  27           3468   Male        R&D                   6                         2       High School
6 STAFF-8  32           3068   Male        R&D                   8                         3 Associates Degree
```

Statistical Summary of Variables:

```
      Age        Monthly.Income      Gender     Department  Total.Working.Years Years.Since.Last.Promotion              Education
 Min.   :18.00   Min.   : 1009   Female:588   HR   : 63   Min.   : 0.00        Min.   : 0.000             High School       :170
 1st Qu.:30.00   1st Qu.: 2911   Male  :882   R&D  :961   1st Qu.: 6.00        1st Qu.: 0.000             Associates Degree:282
 Median :36.00   Median : 4919                Sales:446   Median :10.00        Median : 1.000             Bachelor's Degree:572
 Mean   :36.92   Mean   : 6503                            Mean   :11.28        Mean   : 2.188             Master's Degree  :398
 3rd Qu.:43.00   3rd Qu.: 8379                            3rd Qu.:15.00        3rd Qu.: 3.000             Doctoral Degree  : 48
 Max.   :60.00   Max.   :19999                            Max.   :40.00        Max.   :15.000
```

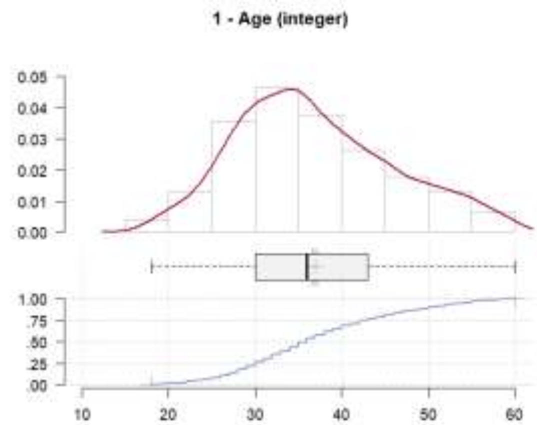
- **Workforce Maturity:** The mean age is approximately **37 years**, with the middle 50% of employees (Interquartile Range) falling between **30** and **43 years**. This suggests a mid-career professional workforce.
- **Income Distribution:** There is a significant gap between the minimum income (**1,009**) and the maximum (**19,999**). Since the Mean **(6,503)** is considerably higher than the Median **(4,919)**, the data is "right skewed," meaning a smaller group of high-level earners is pulling the average upward.
- **Gender Composition:** The dataset reveals a notable gender imbalance, with **60%** of the workforce being Male (**n=882**) and **40%** being Female (**n=588**). This distribution indicates a ratio of approximately **1.5 men for every 1 woman** in the organization.
- **Departmental Balance:** The **Research and Development** department is the largest by far with **961 employees**, nearly doubling the size of Sales (446) and Human Resources (63).
- **Organizational Tenure and Promotion:** While some employees have **40 years** of experience, the median is **10 years**. Interestingly, the average time since the last promotion is quite low (**2.18 years**), though some individuals have gone **15 years** without a promotion, which could be a significant factor in attrition analysis.
- **Educational Attainment:** The workforce is highly educated. **Bachelor's Degrees (572)** and **master's Degrees (398)** make up most of the population, while only a small fraction (**48**) hold Doctoral degrees.

Descriptive Analysis:

<u>Variables Descriptive</u>

- **Variable (1): Age**

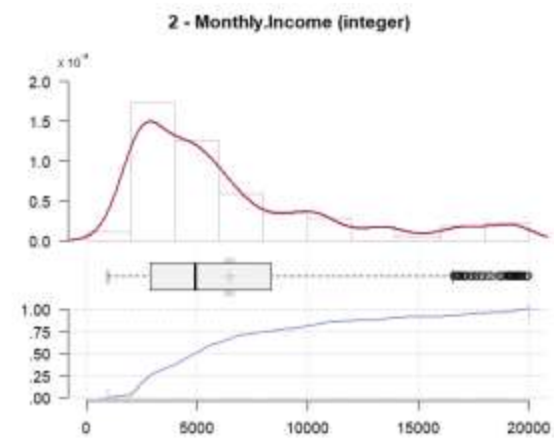
```
1 - Age (integer)

  length      n    NAs  unique     0s   mean  meanCI'
   1'470  1'470      0      43      0  36.92   36.46
         100.0%   0.0%           0.0%          37.39

     .05    .10    .25  median    .75    .90     .95
   24.00  26.00  30.00   36.00  43.00  50.00   54.00

   range     sd  vcoef     mad    IQR   skew    kurt
   42.00   9.14   0.25    8.90  13.00   0.41   -0.41

lowest : 18 (8), 19 (9), 20 (11), 21 (13), 22 (16)
highest: 56 (14), 57 (4), 58 (14), 59 (10), 60 (5)

' 95%-CI (classic)
```



  - The "Age" variable follows a near-normal distribution with a slight positive skew (0.41) and a mean of 36.92 (SD = 9.14).
  - The absence of missing values (NAs) and a 95% confidence interval of (36.46, 37.39) indicate high data reliability.
  - The low Kurtosis (-0.41) suggests a relatively flat distribution compared to a normal curve, confirming representation across most age brackets between the minimum of 18 and maximum of 60.

- **Variable (2): Monthly Income**

```
2 - Monthly.Income (integer)

    length          n       NAs    unique        0s       mean    meanCI'
     1'470      1'470         0     1'349         0   6'502.93   6'262.06
               100.0%      0.0%                0.0%              6'743.80

       .05        .10       .25    median       .75        .90        .95
  2'097.90   2'317.60  2'911.00  4'919.00  8'379.00  13'775.60  17'821.35

     range         sd     vcoef       mad       IQR       skew       kurt
 18'990.00   4'707.96      0.72  3'260.24  5'468.00       1.37       0.99

lowest : 1'009, 1'051, 1'052, 1'081, 1'091
highest: 19'859, 19'926, 19'943, 19'973, 19'999

' 95%-CI (classic)
```

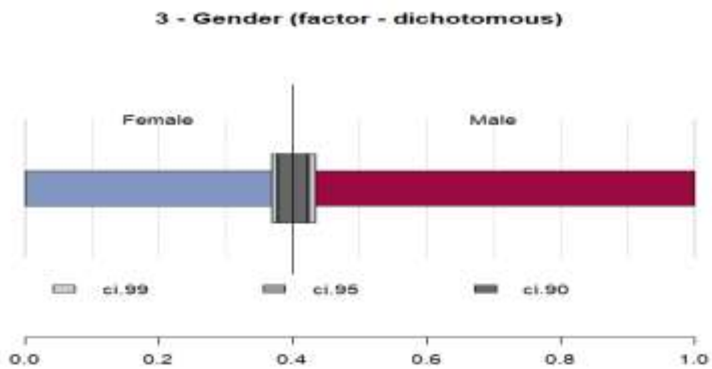


  - The Monthly Income variable is significantly right-skewed (1.37).
  - The distribution is non-normal, with a heavy concentration of values toward the lower end of the scale; the median (4,919) sits well below the mean (6,502.93).
  - The 95% confidence interval (6,262.06, 6,743.80) indicates high precision in the mean estimate.
  - Given the standard deviation of 4,707.96, there is considerable dispersion in the data.

- The boxplot highlights numerous outliers on the high end (above $15,000), representing senior leadership or specialized technical roles. The Cumulative Distribution Function (CDF) shows that roughly 75% of employees earn less than $8,379 per month.

- **Variable (3): Gender**

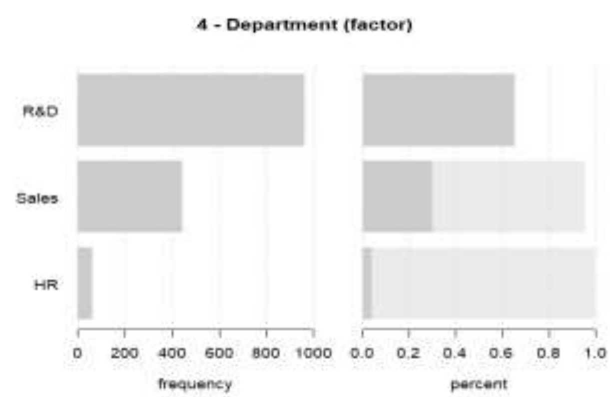
```
3 - Gender (factor - dichotomous)

  length      n    NAs unique
   1'470  1'470      0      2
         100.0%   0.0%

         freq   perc  lci.95  uci.95'
Female    588  40.0%   37.5%   42.5%
Male      882  60.0%   57.5%   62.5%

' 95%-CI (Wilson)
```



  - The Gender variable is a factor with no missing observations.
  - Statistical analysis reveals that the proportion of Male employees (60.0%) is significantly higher than that of Female employees (40.0%).
  - According to the Wilson 95% confidence intervals, the true population proportion for Female employees likely falls between 37.5% and 42.5%, confirming a consistent gender imbalance toward male representation within the organization.

- **Variable (4): Department**

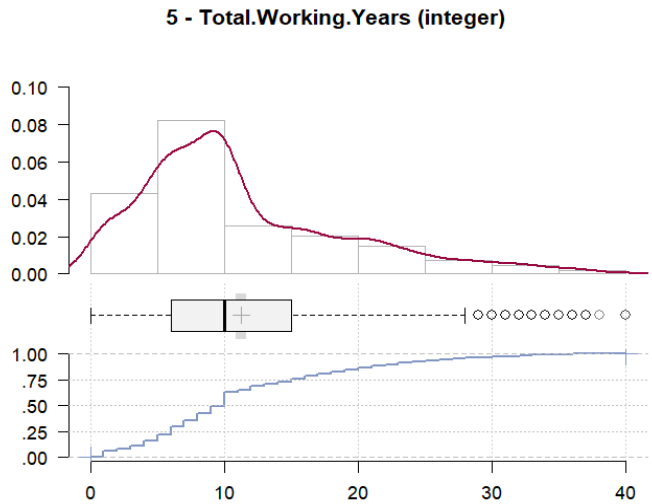
```
4 - Department (factor)

  length      n    NAs unique levels  dupes
   1'470  1'470      0      3      3      y
         100.0%   0.0%

    level  freq   perc  cumfreq  cumperc
1     R&D   961  65.4%      961    65.4%
2   Sales   446  30.3%    1'407    95.7%
3      HR    63   4.3%    1'470   100.0%
```



  - The “Department”factor illustrates a high degree of departmental concentration.
  - With a cumulative percentage of 95.7% shared between R&D and Sales, the organization’s primary resources are focused on product innovation and revenue generation.
  - The small size of the HR department (4.3%) indicates a high employee-to-HR ratio.
  - The density plot and boxplot confirm a near-normal distribution with a slight positive skew (0.41), showing a healthy concentration of talent in their peak career years.

- **Variable (5): Total Working Year**

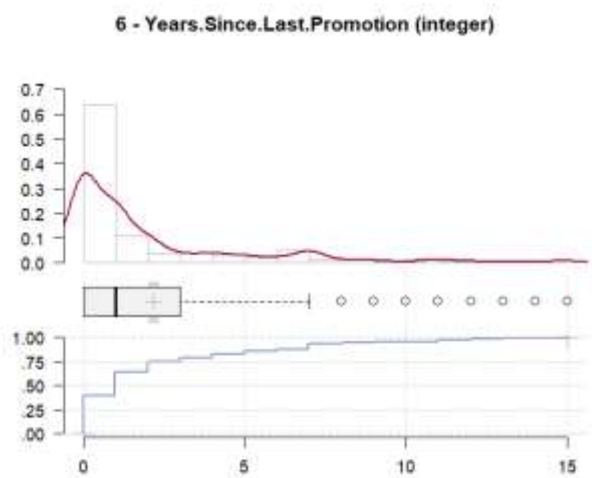
```
5 - Total.Working.Years (integer)

  length       n     NAs  unique      0s    mean  meanCI'
   1'470   1'470       0      40      11   11.28   10.88
          100.0%    0.0%            0.7%           11.68

     .05     .10     .25  median     .75     .90     .95
    1.00    3.00    6.00   10.00   15.00   23.00   28.00

   range      sd   vcoef     mad     IQR    skew    kurt
   40.00    7.78    0.69    5.93    9.00    1.11    0.91

lowest : 0 (11), 1 (81), 2 (31), 3 (42), 4 (63)
highest: 35 (3), 36 (6), 37 (4), 38, 40 (2)

heap(?): remarkable frequency (13.7%) for the mode(s) (= 10)

' 95%-CI (classic)
```



   - The “Total Working Years” variable follows a non-normal distribution with a positive skew of 1.11 and a kurtosis of 0.91.
   - The presence of a statistical “heap” at the 10-year mark (representing 13.7% of observations) suggests a specific recruitment profile or industry-standard experience level for most hires.
   - The interquartile range (IQR) of 9 years and a standard deviation of 7.78 reflect moderate variability in career tenure. The 95% confidence interval (10.88, 11.68) indicates that the mean experience level is a highly reliable metric for this population.

- **Variable (6): Years Since Last Promotion**

```
6 - Years.Since.Last.Promotion (integer)

  length       n     NAs  unique      0s    mean  meanCI'
   1'470   1'470       0      16     581    2.19    2.02
          100.0%    0.0%           39.5%            2.35

     .05     .10     .25  median     .75     .90     .95
    0.00    0.00    0.00    1.00    3.00    7.00    9.00

   range      sd   vcoef     mad     IQR    skew    kurt
   15.00    3.22    1.47    1.48    3.00    1.98    3.59

lowest : 0 (581), 1 (357), 2 (159), 3 (52), 4 (61)
highest: 11 (24), 12 (10), 13 (10), 14 (9), 15 (13)

heap(?): remarkable frequency (39.5%) for the mode(s) (= 0)

' 95%-CI (classic)
```

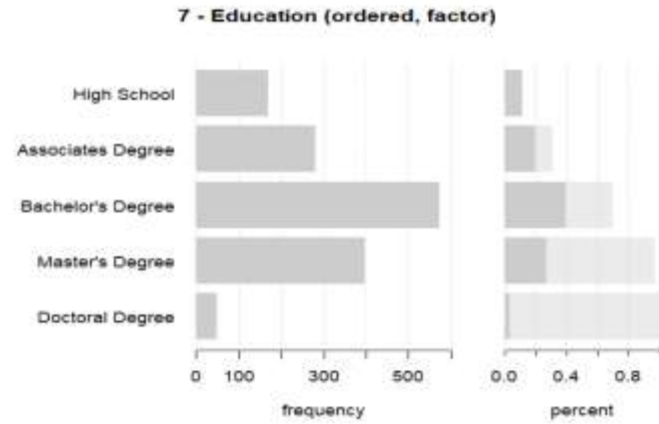


   - The distribution for “Years Since Last Promotion” is heavily right-skewed (1.98) .
   - 1,470 observations, significantly pulling the median (1.00) below the mean (2.19).
   - The standard deviation of 3.22 and a high coefficient of variation (1.47) indicate substantial dispersion in the timing of career advancement across the company.
   - The 95% confidence interval (2.02, 2.35) confirms the reliability of the average promotion cadence within this sample.

- The "Years Since Last Promotion" density plot is heavily right-skewed (1.98).

- **Variable (7): Education**

```
7 - Education (ordered, factor)

  length      n    NAs unique levels   dupes
   1'470  1'470      0      5      5       y
        100.0%    0.0%

                 level  freq   perc  cumfreq  cumperc
1          High School   170  11.6%      170    11.6%
2   Associates Degree    282  19.2%      452    30.7%
3   Bachelor's Degree    572  38.9%    1'024    69.7%
4     Master's Degree    398  27.1%    1'422    96.7%
5     Doctoral Degree     48   3.3%    1'470   100.0%
```



  - The distribution is centered on Level 3 (Bachelor's Degree), which serves as the mode for the dataset.
  - Cumulative data shows that (69.7%) of employees have an educational level of Bachelor's or below, while nearly a third of the workforce (30.4%) has achieved post-graduate status (Master's or Doctoral degrees).
  - There are zero missing values (NAs), indicating high data integrity for this demographic marker.

Summary of Qualitative Indicators

Relative Frequency

1. Education:

Education Level	Relative Frequency	Cumulative Relative Frequency
High School	0.1156	0.1156
Associates Degree	0.1918	0.3075
Bachelor's Degree	0.3891	0.6966
Master's Degree	0.2707	0.9673
Doctoral Degree	0.0327	1.0000

2. Department:

Department	Relative Frequency	Cumulative Relative Frequency
HR	0.0429	0.0429
R&D	0.6537	0.6966
Sales	0.3034	1.0000

3. Gender**:**

Gender	Relative Frequency	Cumulative Relative Frequency
Female	0.4000	0.4000
Male	0.6000	1.0000

Mode:

Category	Mode	Relative Frequency
Gender	Male	0.60
Department	R&D	0.65
Education	Bachelor's Degree	0.39

The organizational structure is primarily driven by technical innovation, with the R&D department serving as the central hub of operations (65.4% of headcount).

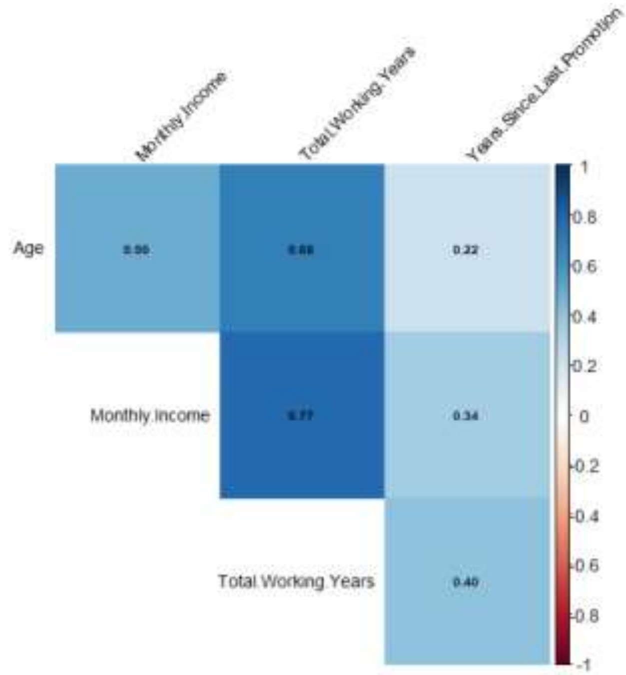
The workforce is highly specialized, characterized by a dominant Bachelor's Degree academic profile (38.9%) and significant post-graduate representation.

While the gender split is 60% Male and 40% Female, the lean nature of the HR department (4.3%) suggests an efficient, product-focused operational model.

Correlation Analysis for Numeric Variables

Strongest Relationship: Monthly Income and Total Working Years (0.77)

- This is the strongest positive correlation in the matrix.
- **Meaning:** As an employee's total working years increase, their monthly income tends to increase significantly. This suggests that experience is a primary driver of pay in this dataset.



Strong Relationship: Age and Total Working Years (0.68)

- This is logically expected. As people get older, they generally accumulate more years of work experience.

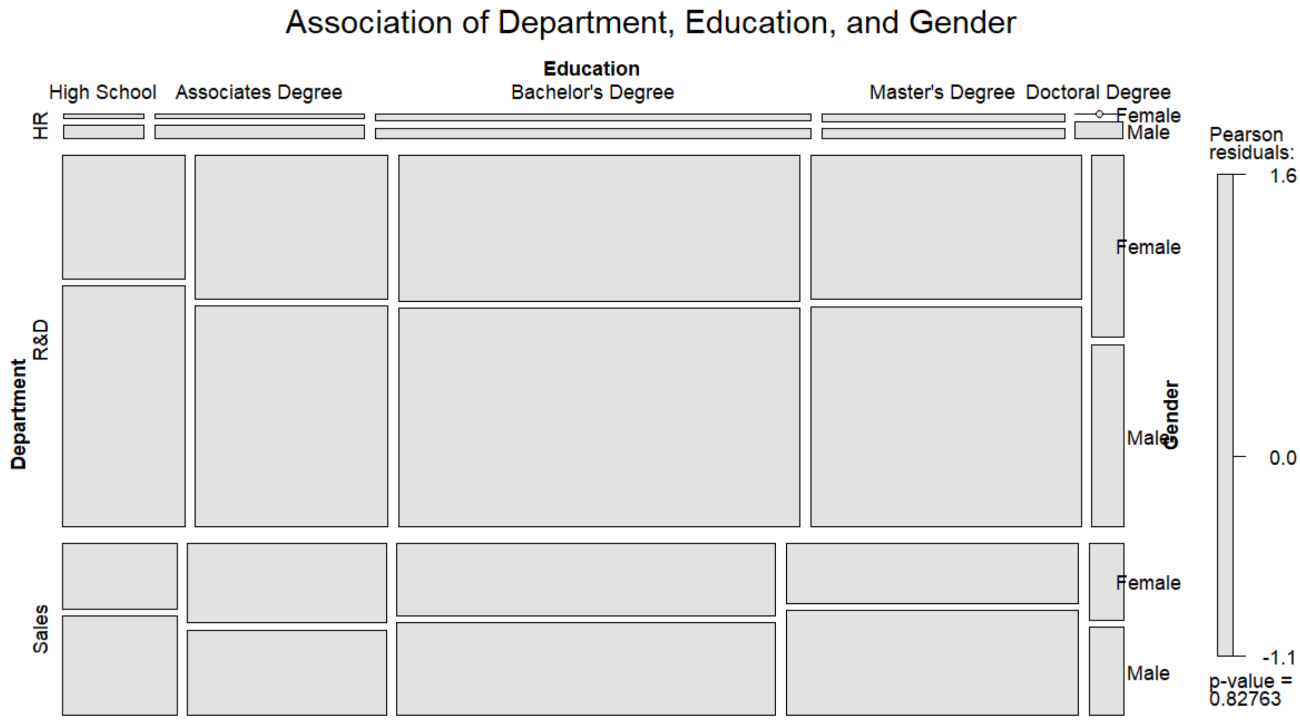
Moderate Relationship: Age and Monthly Income (0.50)

- There is a clear positive correlation, but it is weaker than the correlation between income and working years.

Weakest Relationships: Years Since Last Promotion

- This variable has the lowest correlation with all others (ranging from 0.22 to0.40).
- The highest correlation for this category is with Total Working Years (0.40), suggesting that those who have been in the workforce longer might also have longer gaps between promotions.

Mosaic plot: Association between Department, Education, and Gender



Dominant Department: The R&D department is significantly larger than Sales or HR across almost all education levels.

Common Education Levels: Bachelor's and Master's Degrees represent the vast majority of the workforce. Doctoral Degrees and High School diplomas represent the smallest portions.

Gender Balance: Within the large R&D and Sales blocks, the split between Male and Female appears relatively consistent, though there is a slight visual lean toward more males in some R&D segments.

HR Presence: The HR department is extremely small relative to the others, appearing as thin slivers at the top of the plot.

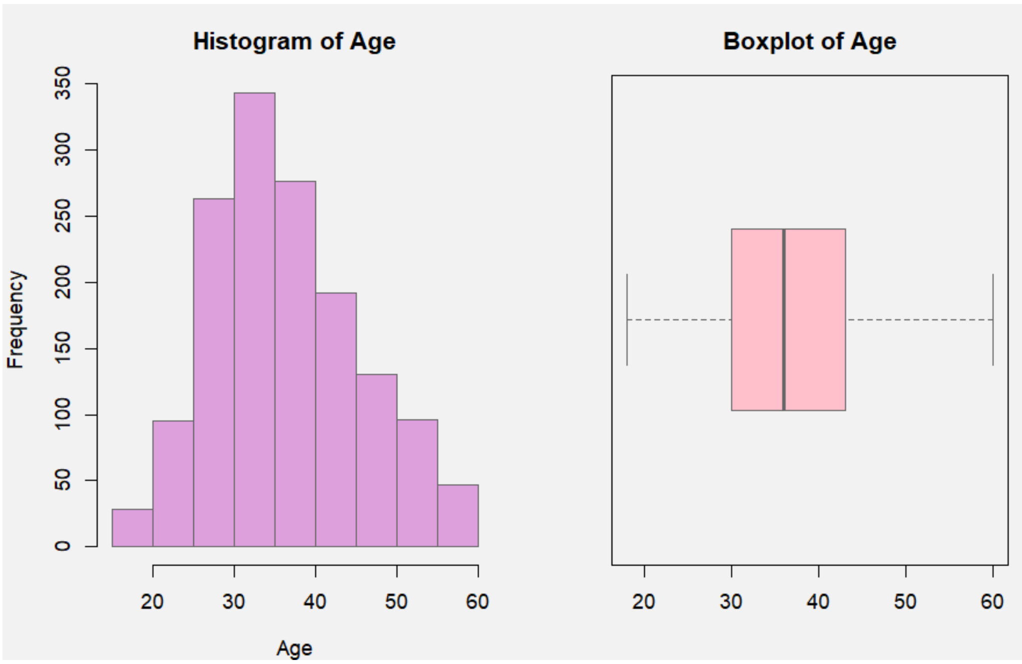
P-value of 0.82763: This high p-value suggests that the distribution of Gender and Education is relatively "independent" of the Department.

Pearson residuals: (indicated by the grey color scale) confirm this. If a box were colored deep blue or deep red, it would mean that specific group is significantly over- or under-represented. Since all boxes are grey, the distribution is considered "typical."

Exploratory Data Visualization (FUNCTION):

1. **Age:**

The distribution of the "Age" variable was evaluated using a histogram and a boxplot to identify central tendency, spread, and shape.



1. Distribution Shape and Skewness

- **Right-Skewed Distribution:** The histogram shows a distribution that is moderately **right-skewed** (positively skewed). There is a higher concentration of individuals in the younger age brackets, specifically between **30 and 40 years old**, with the frequency gradually tapering off as age increases toward 60.
- **Unimodal Pattern:** The data is unimodal, with the peak frequency (the mode) occurring in the **30–35 age bin**, where the count exceeds 300 observations.

2. Central Tendency and Spread

- **Median Age:** The boxplot indicates a median age of approximately **36 years**, represented by the thick vertical line within the interquartile range (IQR) box.
- **Interquartile Range:** The bulk of the middle 50% of the population falls between roughly **30 and 43 years of age**.
- **Total Range:** The dataset captures a diverse age range starting from approximately **18 years** up to **60 years**.

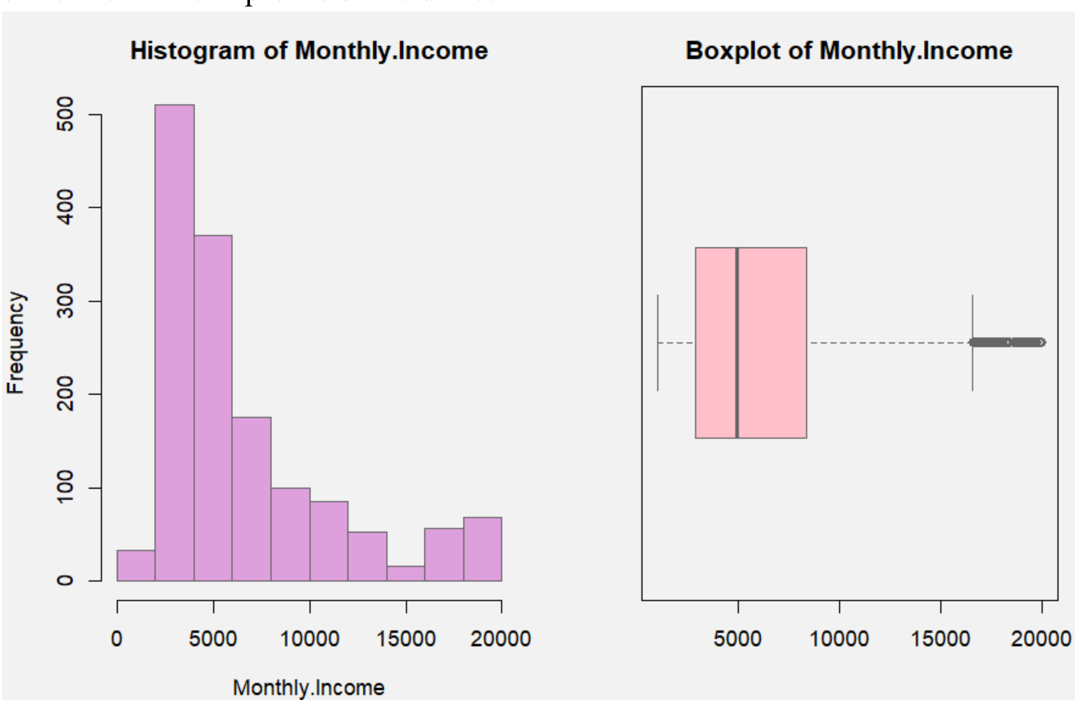
3. Outliers and Observations

- **No Significant Outliers:** The boxplot does not show any individual points (dots) beyond the whiskers, suggesting that there are no extreme statistical outliers in this dataset. The "whiskers" extend to the minimum and maximum values (approx. 18 and 60) without indicating data points that are abnormally far from the rest of the distribution.

- The sample is primarily composed of young-to-middle-aged adults.
- The slight right skew suggests that while the population is relatively young, there is a consistent representation of older individuals up to age 60, without any extreme outliers skewing the results.

2. Monthly Income

The distribution of "Monthly Income" was analyzed using a histogram and a boxplot to understand the financial profile of the dataset.



1. Distribution Shape and Skewness

- **Strong Positive Skew:** The histogram reveals a distribution that is heavily **right-skewed** (positively skewed).
- **Concentration of Wealth:** The vast majority of the population earns a monthly income at the lower end of the scale, specifically between **$2,000 and $6,000**.
- **Long Tail:** There is a significant "long tail" extending toward the higher income brackets, with frequency dropping sharply as income exceeds $10,000.

2. Central Tendency and Variation

- **Median Income:** The boxplot indicates a median monthly income of approximately **$5,000**.
- **Interquartile Range (IQR):** The middle 50% of the sample earns between roughly **$3,000 and $8,500** per month.
- **Income Gap:** The significant distance between the median and the maximum values highlights a high degree of income inequality within the group.

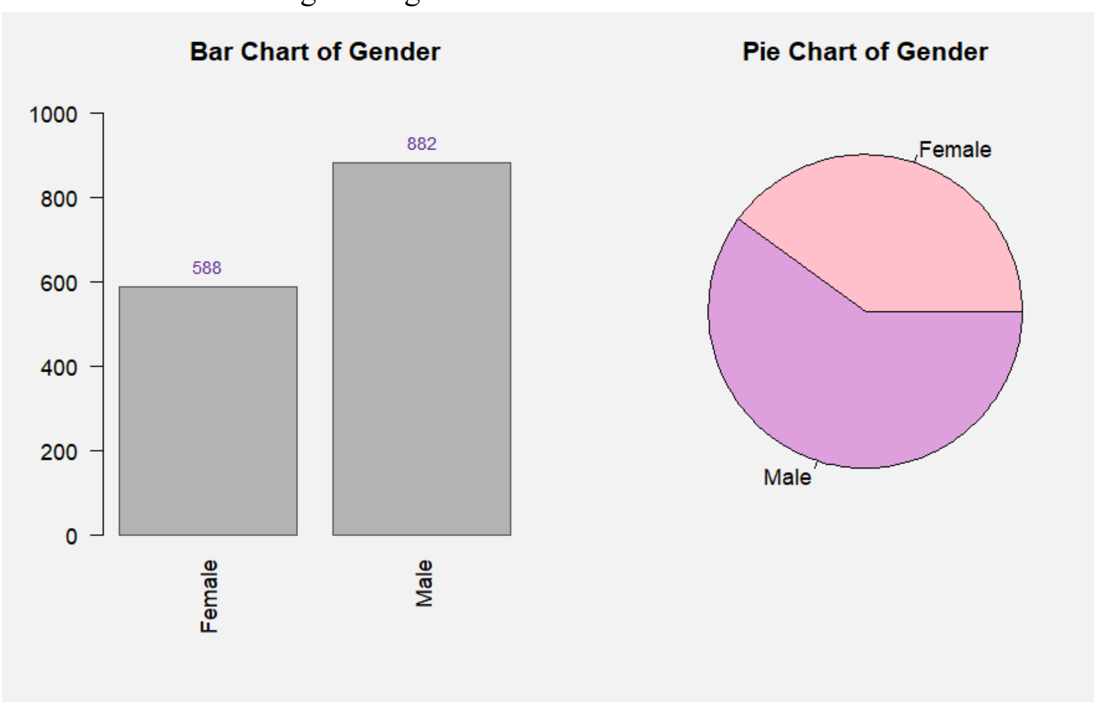
3. Outliers and Extremes

- **Presence of High-Income Outliers:** Unlike the age distribution, the monthly income boxplot displays several **statistical outliers**.
- **Upper Extremes:** These outliers represent a segment of "high earners" who make between **$17,000 and $20,000** per month. These individuals sit well above the upper whisker, indicating they are mathematically distinct from the general population trend.

➢ The dataset is characterized by a "low-income" majority, with the most common earners falling in the $2,500–$5,000 range. However, the presence of a small group of high-earning outliers pulls the mean upward, suggesting that the **median** is a more accurate representation of the "typical" individual's income than the average would be

3. Gender

The distribution of " Gender" was analyzed using a barchart and a pie chart to understand the spread of the dataset according to the gender



1. Distribution and Proportionality

- **Male Dominance:** The data reveals a distribution that is moderately skewed toward male participants.
- **Dominant Group:** The bar chart indicates that males are the primary demographic, with a frequency of 882, compared to 588 females.
- **Composition Ratio:** The pie chart confirms that the male segment occupies a larger share of the total, resulting in a 60/40 percentage split.

2. Comparative Frequency and Volume

- **Frequency Gap:** There is a clear disparity in volume; the male count is exactly 1.5 times greater than the female count.
- **Sample Size Impact:** With a total sample size of 1,470, the difference of 294 individuals between the two groups represents a significant margin that could influence gender-specific correlations in further analysis.
- **Visual Consistency:** Both visual representations (Bar and Pie) demonstrate a consistent 3:2 ratio, highlighting a stable imbalance across the dataset.

3. Statistical Representation

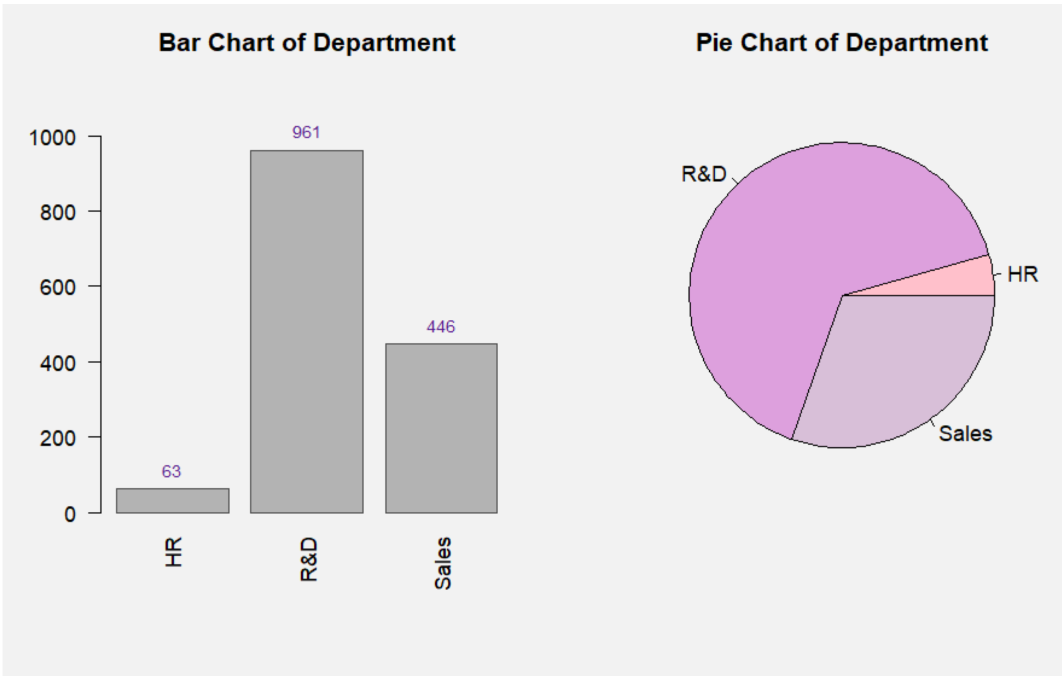
- **Majority vs. Minority:** The dataset is characterized by a "Male-majority" structure. This suggests that any aggregate averages calculated for the whole group will be more heavily weighted by male data points.
- **Demographic Weight:** While the female representation is substantial (40%), it sits significantly below the male baseline, indicating that the population is not evenly balanced for a 1:1 gender comparison.

➢ The gender distribution is characterized by a **60% male majority**.

➢ While the distribution is not as extreme as the income skewness, the 20-percentage-point gap between the two groups suggests that gender should be treated as a controlled variable in any subsequent performance or behavioral modeling.

4. Department

The distribution of "Department" was analyzed using a bar chart and a pie chart to understand the organizational structure and workforce allocation across the dataset.



1. Distribution Shape and Heavy Concentration

- **Dominant Departmental Focus:** The bar chart reveals a distribution that is heavily concentrated within a single functional area, Research & Development (R&D).
- **Organizational Core:** Most of the workforce is in R&D, which accounts for **961** individuals.
- **Steep Decline:** There is a sharp drop in frequency when moving from R&D to the Sales and HR departments, indicating a highly specialized or operationally focused organizational structure.

2. Proportional Representation and Variation

- **Primary Majority:** The pie chart illustrates that R&D comprises approximately **65%** of the total population, making it the clear majority stakeholder.
- **Secondary Sector:** Sales represents the second largest group with **446** members, or roughly **30%** of the workforce.
- **Departmental Gap:** The significant distance between the R&D count (961) and the Sales count (446) highlights a massive resource allocation toward technical or developmental roles compared to commercial ones.

3. Outliers and Minorities

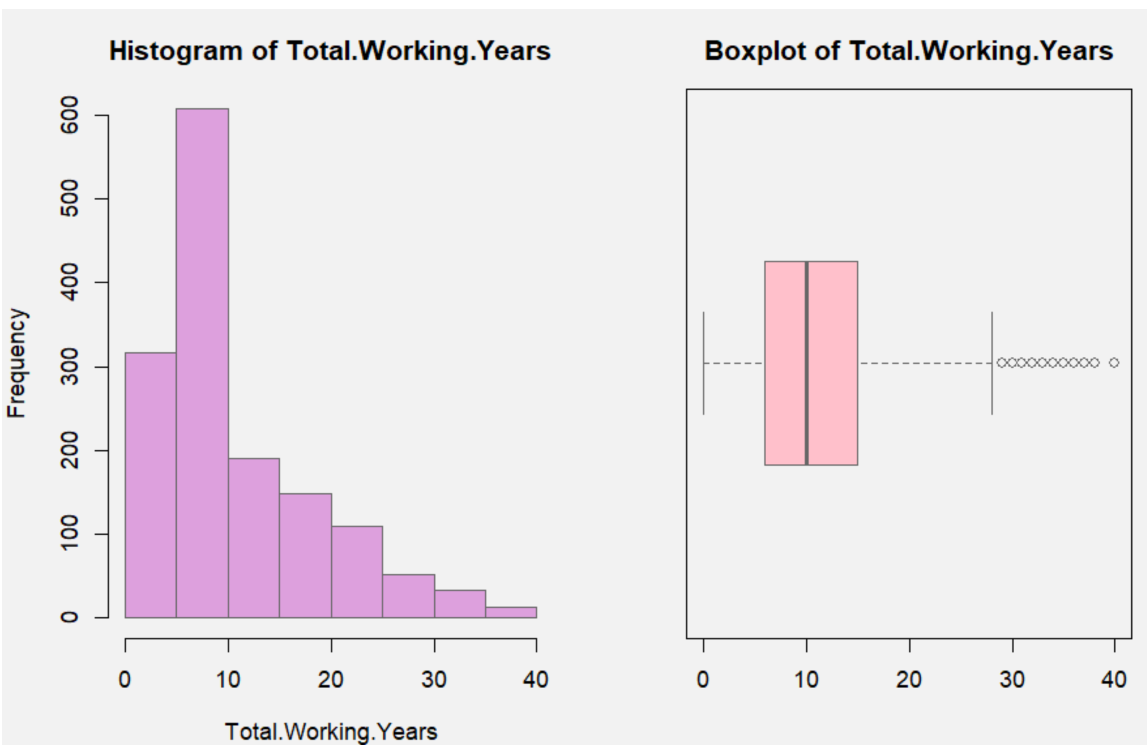
- **Presence of a "Micro-Department":** Unlike the relatively balanced split seen in other demographics, the HR department serves as a distinct minority.
- **Lower Extremes:** With only **63** individuals, the HR department sits well below the other two categories, representing only about **4%–5%** of the total organization.
- **Statistical Distinctness:** These individuals are mathematically distinct from the general population trend, functioning as a small support pillar in an R&D-heavy environment.

➢ The dataset is characterized by an **"R&D-centric"** majority, with nearly two-thirds of all employees falling into this single category.

➢ The presence of a small HR group and a mid-sized Sales team suggests that the R&D department will heavily influence the "typical" employee profile, and any company-wide findings will be primarily driven by the behaviors and metrics of the R&D staff.

5. Total Working Years

The distribution of "Total Working Years" was analyzed using a histogram and a boxplot to understand the professional experience level of the dataset.



1. Distribution Shape and Skewness

- **Moderate Positive Skew:** The histogram reveals a distribution that is right-skewed, indicating that most of the population consists of early-to-mid-career professionals.
- **Concentration of Experience:** The highest frequency of individuals is concentrated in the **5–10-year** range, which serves as the primary mode of the dataset.
- **Tapered Tail:** There is a visible "tail" extending toward the more senior brackets, with the frequency of individuals steadily declining as total working years exceed **20 years**.

2. Central Tendency and Variation

- **Median Experience:** The boxplot indicates a median of approximately **10 total working years**.
- **Interquartile Range (IQR):** The middle 50% of the sample possesses between roughly **6 and 15 years** of experience.
- **Professional Diversity:** The spread from 0 to nearly 30 years (within the whiskers) highlights a wide variety of experience levels, though the bulk of the workforce remains relatively young in their careers.

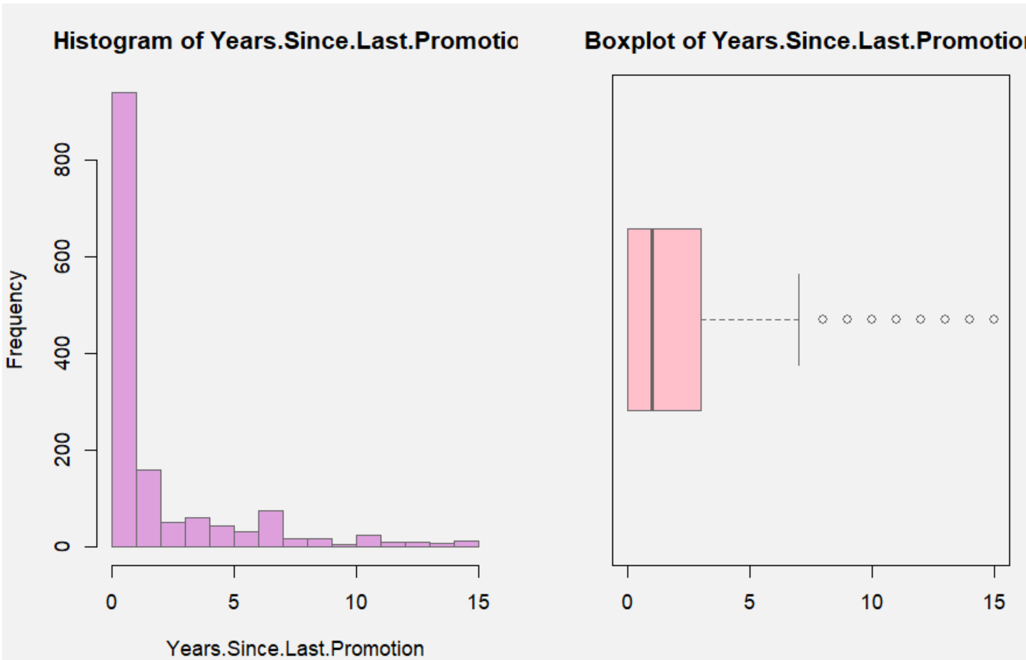
3. Outliers and Extremes

- **Presence of Senior Outliers:** Like the income distribution, the boxplot for total working years displays several statistical outliers at the upper end of the scale.
- **Upper Extremes:** These outliers represent a small segment of highly experienced "veterans" who have worked between **30 and 40 years**.
- **Statistical Distinction:** These individuals sit beyond the upper whisker, indicating their level of experience is mathematically distinct from the broader workforce trend.

➢ The dataset is characterized by a **"mid-career" majority**, with the most common experience level falling to under 10 years. While the distribution includes a small group of senior outliers, the median of 10 years provides a more accurate representation of the "typical" employee's professional tenure than an average influenced by the long tail of veterans.

6. Years Since Last Promotion

The distribution of "Years Since Last Promotion" was analyzed using a histogram and a boxplot to understand the promotional dynamics and career progression patterns within the dataset.



1. Distribution Shape and Skewness

- **Extreme Positive Skew:** The histogram reveals a heavily right-skewed distribution, with an overwhelming concentration of observations at zero years since last promotion, indicating that most employees have been recently promoted or are new to their current role.
- **Dominant Zero Point:** The most striking feature is the extraordinarily high frequency at 0 years (900+ individuals), which serves as the absolute mode and represents nearly the entire sample's central mass.
- **Elongated Tail:** Beyond the initial spike, there is a steep drop-off in frequency, with a long, thin tail extending toward 15 years, suggesting a small subset of employees who have experienced prolonged periods without advancement.

2. Central Tendency and Variation

- **Low Median Tenure:** The boxplot indicates a median of approximately 1 year since last promotion, reflecting the organization's pattern of frequent promotional activity or recent workforce influx.
- **Compressed Interquartile Range (IQR):** The middle 50% of employees fall within an extremely narrow range of roughly 0 to 2 years, demonstrating limited variation in promotion recency for the bulk of the workforce.

- **Minimal Typical Variation:** The tight clustering suggests that the "typical" employee experience is one of recent promotion or role change, with little differentiation among the majority.

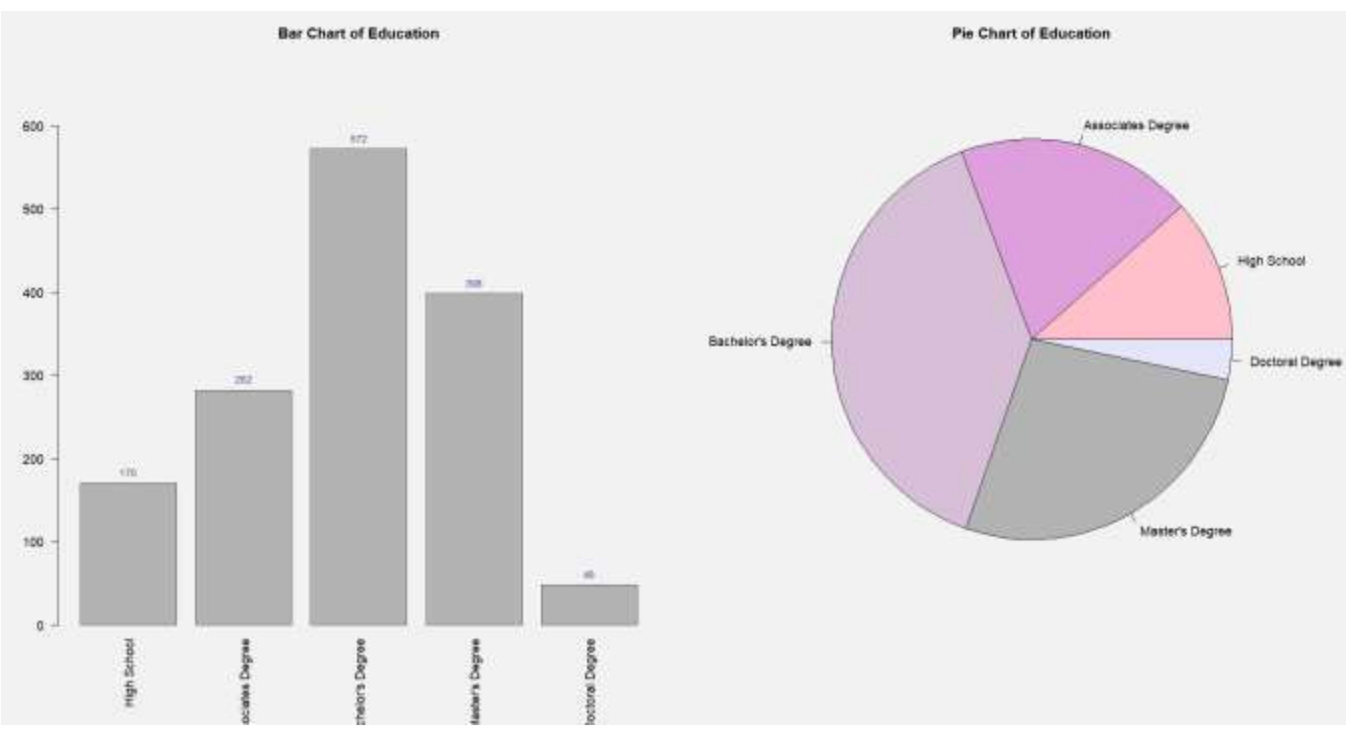
3. Outliers and Extremes

- **Substantial Outlier Population:** The boxplot displays numerous statistical outliers beginning around 7 years and extending to 15+ years since last promotion.
- **Stagnation Indicators:** These outliers represent individuals who have remained in their current positions for extended periods, potentially signaling career stagnation, specialized roles with limited advancement paths, or employees who have plateaued professionally.
- **Statistical Separation:** The presence of so many outliers beyond the upper whisker indicates a clear bifurcation between the recently promoted majority and a distinct minority experiencing prolonged promotion gaps.

➢ The dataset reveals a promotional landscape dominated by recent advancement activity, with the median employees having received their last promotion within just one year. However, the substantial number of outliers suggests a dual workforce structure: a large, actively progressing majority alongside a smaller cohort experiencing significant promotional delays that may warrant organizational attention regarding career development and retention strategies.

7. Education

The distribution of "Education" was analyzed using a bar chart and a pie chart to understand the educational qualifications and academic composition of the workforce within the dataset.



1. Distribution Shape and Concentration

- **Extreme Positive Skew:** The bar chart reveals a heavily imbalanced distribution, with Research & Development (R&D) dominating the organizational landscape and representing most of the workforce.
- **Concentration in R&D:** The highest frequency of individuals is concentrated in R&D (961 employees, representing approximately 65% of the workforce), which serves as the dominant mode of the dataset and defines the organization's core operational identity.
- **Steep Decline Pattern:** There is a dramatic drop in frequency when moving from R&D to Sales (446 employees) and an even sharper decline to HR (63 employees), indicating a highly specialized, innovative-driven organizational structure rather than a balanced multi-functional enterprise.

2. Central Tendency and Variation

- **R&D-Centered Organization:** Both visualizations indicate that the median departmental affiliation falls squarely within R&D, reflecting an organization that prioritizes technical innovation and product development as its primary business function.
- **Limited Functional Diversity:** The concentration across only three departments with extreme variation (65%, 30%, and 4% respectively) demonstrates a narrow organizational focus, with most resources allocated to technical rather than commercial or administrative functions.
- **Secondary Commercial Presence:** Sales represent a substantial but subordinate function (approximately 30% of the workforce), suggesting the organization maintains meaningful customer-facing operations while keeping its primary investment in research and development activities.

3. Departmental Extremes and Minority Groups

- **Minimal Administrative Support:** The dataset shows only 63 employees (approximately 4%) in HR, indicating that administrative and people management functions operate with minimal staffing, potentially reflecting a high employee-to-HR ratio or lean support infrastructure.
- **R&D-Sales Divide:** The 2:1 ratio between R&D and Sales personnel highlights a clear strategic prioritization of innovation over commercial expansion, typical of technology companies, research institutions, or product-focused enterprises in growth phases.
- **Functional Imbalance:** The pie chart clearly illustrates that R&D alone accounts for nearly two-thirds of all personnel, with Sales and HR combined representing only 35%, creating a top-heavy organizational structure weighted toward technical capabilities.

➢ The dataset is characterized by an "R&D-centric" organizational model, with most employees concentrated in technical and development functions.

- The distribution reveals a specialized workforce where 65% of all personnel work in research and development, suggesting either a technology startup, pharmaceutical company, engineering firm, or innovation-focused enterprise. The moderate Sales presence (30%) indicates commercial operations exist but remain secondary to technical priorities, while the minimal HR function (4%) suggests lean administrative support.
- This departmental composition means that any organization-wide analysis will be heavily influenced by R&D employee characteristics, behaviors, and performance metrics, potentially masking distinct patterns within the smaller Sales and HR populations.
- The structure reflects a company where innovation and technical excellence constitute the core competitive advantage and primary value proposition.

Regression Model:

- **To what extent do demographic, educational, and professional factors collectively predict an employee's monthly income, and which of these factors serves as the most influential driver of compensation?**

Multiple linear regression is an appropriate and effective analytical method for this research because the dependent variable, **monthly income**, is continuous and is influenced by **multiple predictors simultaneously**. The model allows for the estimation of the **independent contribution** of demographic (e.g., age, gender), educational (education level), and professional or organizational factors (total working years, years since last promotion, department) while **controlling for the effects of other variables**.

Moreover, multiple regression enables the comparison of **standardized coefficients**, which helps identify **the most influential driver of compensation** among the predictors. By estimating the model on separate data subsets, the analysis strengthens **model validity**, supports **diagnostic checks** (such as multicollinearity and residual behavior), and improves the **generalizability of the findings**. Overall, this approach provides a comprehensive framework for understanding both the **collective and individual effects** of employee characteristics on income.

i. **Initial Linear Model:**

Monthly Incomei=β0+β1(Agei)+β2(Total Working Yearsi)+β3(Genderi)+β4 (Years Since Last Promotioni)+β5 (DepartmentR&D_i)+ β6(DepartmentSales_i)+β7(EducationBachelors_Degree_i)+β8(EducationDoctoral_Degree_i)+ β9 (Education. High_School_i) **+ β10(EducationMasters_Degree_i) + εi**

Where the dummy variables are defined as:

$$\textbf{GenderMale}_i = \begin{cases} \mathbf{1} & \textbf{if employee is male} \\ \mathbf{0} & \textbf{if employee is female (reference)} \end{cases}$$

$$\textbf{DepartmentR\&D}_i = \begin{cases} \mathbf{1} & \textbf{if employee is in R\&D} \\ \mathbf{0} & \textbf{otherwise} \end{cases}$$

$$\textbf{DepartmentSales}_i = \begin{cases} \mathbf{1} & \textbf{if employee is in Sales} \\ \mathbf{0} & \textbf{otherwise (reference = HR)} \end{cases}$$

$$\textbf{EducationBachelors}_i, \textbf{EducationMasters}_i, \textbf{EducationDoctoral}_i, \textbf{High School} = \begin{cases} \mathbf{1} & \textbf{if employee has that degree} \\ \mathbf{0} & \textbf{otherwise (reference = EducationAssociates}_i\textbf{)} \end{cases}$$

ε_i**is the residual error term for employee i.**

The fitted model:

$\widehat{\text{Monthly Income}}$_ı = 2068.88 – 26.42 (Agei) + 478.52 (Total Working Yearsi)- 35.38(GenderMalei)+ 50.69 (Years Since Last Promotioni) - 490.75 DepartmentR\&Di) + 314.33 (DepartmentSales)i + 197.19 (Education Bachelor_i)+ 927.24 (Education. Doctoral_i)+ 512.27 (Education. High_School_i) +35.18 (Education Master_i) .

- **Testing The Whole Model Significance Using F-test :**

H0: β1=β2=…….= β10 = 0 V.s H1:At least one βj ≠0

Since $F(10, 1018) = 138.1$, $p < 0.05$, indicates strong evidence against the null hypothesis that all

```
Call:
lm(formula = Monthly.Income ~ Age + Total.Working.Years + Gender +
    Years.Since.Last.Promotion + Department + edureg, data = train_data)

Residuals:
    Min      1Q  Median      3Q     Max
-9669.8 -1682.4  -142.8  1340.6 10757.6

Coefficients:
                           Estimate Std. Error t value Pr(>|t|)
(Intercept)                 2068.88     692.12   2.989  0.00286 **
Age                          -26.42      14.18  -1.864  0.06263 .
Total.Working.Years          478.52      17.74  26.977  < 2e-16 ***
GenderMale                   -35.38     194.93  -0.181  0.85602
Years.Since.Last.Promotion    50.69      32.66   1.552  0.12096
DepartmentR&D               -490.75     495.15  -0.991  0.32186
DepartmentSales              314.33     512.54   0.613  0.53983
eduregBachelor's Degree      197.19     258.47   0.763  0.44569
eduregDoctoral Degree        927.24     543.09   1.707  0.08806 .
eduregHigh School            512.27     354.11   1.447  0.14830
eduregMaster's Degree         35.18     276.96   0.127  0.89895
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3037 on 1018 degrees of freedom
Multiple R-squared:  0.5757,    Adjusted R-squared:  0.5715
F-statistic: 138.1 on 10 and 1018 DF,  p-value: < 2.2e-16
```

regression coefficients are zero. Thus, the model is statistically significant, confirming that at least one predictor meaningfully contributes to explaining income variation.

Adjusted R^2 = 0.5715 (~57.2%)

The model explains a substantial portion of the variation in Monthly Income, with Adjusted R^2 = 0.572. This indicates that approximately 57% of the variability in employee income can be accounted for by the predictors included in the model, while the remaining variability is attributed to factors not captured by the model or random variation.

Residuals:

The residuals range widely from -9,669.8 to 10,757.6, with a median near zero, indicating that the model is approximately unbiased on average. However, the asymmetry and extreme values suggest skewness and non-constant variance (heteroscedasticity)

Coefficients' Interpretation:

- **Intercept ($\hat{\beta}_0$ = 2068.88, p = 0.0029)**

Statistically significant. Represents the expected monthly income for a **female employee** working in the **HR department** with an **Associate Degree**, when **Age = 0**, **Total Working Years = 0**, and **Years Since Last Promotion = 0**. This serves as the baseline reference point.

- **Total Working Years($\hat{\beta}_2$ = 478.52, p < 0.001)**

Statistically significant. Each additional year of total work experience is associated with an increase of approximately **478.52 units** in monthly income, holding all other variables constant. This is the **most influential predictor** in the model.

Marginally significant predictor

- **Age ($\hat{\boldsymbol{\beta}}_{\mathbf{1}} = -\mathbf{26.42}, \boldsymbol{p} \approx \mathbf{0.063}$)**

Indicates a slight decrease in monthly income as age increases, after controlling for total working experience. The effect is weak and only significant at the 10% level.

Non-significant predictors ($\hat{\boldsymbol{\beta}}_{\mathbf{3}}, \hat{\boldsymbol{\beta}}_{\mathbf{4}}, \hat{\boldsymbol{\beta}}_{\mathbf{5}}, \hat{\boldsymbol{\beta}}_{\mathbf{6}}, \hat{\boldsymbol{\beta}}_{\mathbf{7}}, \widehat{\boldsymbol{\beta}\mathbf{8}}, \widehat{\boldsymbol{\beta}}_{\mathbf{9}}, \hat{\boldsymbol{\beta}}_{\mathbf{10}}$)

- **GenderMale ($\hat{\beta}_3$ = −35.38, p = 0.856)**

Not statistically significant. Male employees do not earn significantly different monthly income compared to female employees, holding other factors constant.

- **Years Since Last Promotion ($\hat{\beta}_4$ = 50.69, p = 0.121)**

Not statistically significant. Time since last promotion does not have a reliable effect on monthly income once other variables are controlled for.

- **DepartmentR&D ($\hat{\beta}_5$ = −490.75, p = 0.322)**

Not statistically significant. Employees in R&D do not earn significantly different income compared to those in HR.

- **DepartmentSales ($\hat{\beta}_6$ = 314.33, p = 0.540)**

Not statistically significant. Employees in Sales do not earn significantly different income compared to those in HR.

- **Bachelor's Degree (β7 = 197.19, p = 0.446)**

Not statistically significant. Income does not differ significantly from Associate Degree holders.

- **Master's Degree (β8 = 35.18, p = 0.899)**

Not statistically significant. No meaningful income difference relative to Associate Degree holders.

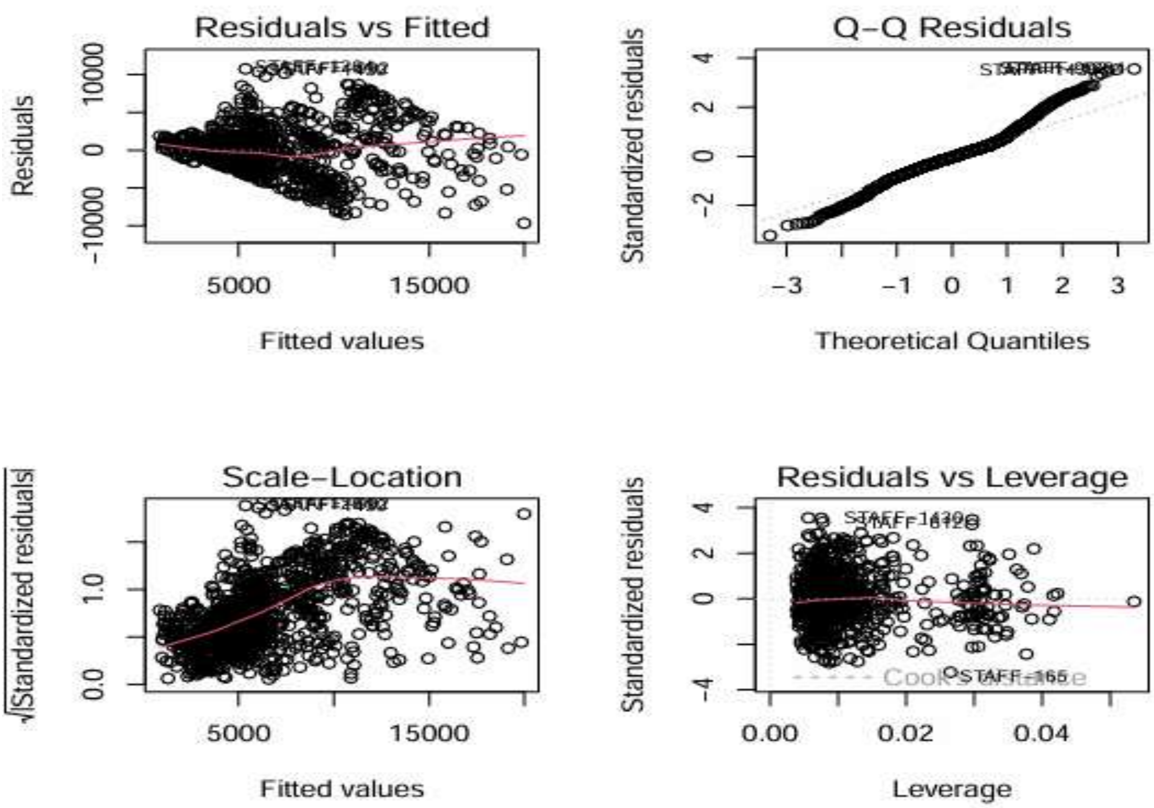
- **High School (β9= 512.27, p = 0.148)**

Not statistically significant. Income does not differ significantly from the reference category.

- **Doctoral Degree (β10 = 927.24, p = 0.088)**

Marginally significant. Employees with a doctoral degree tend to earn higher income than Associate Degree holders, although the effect is weak at the 5% significance level.

Model Diagnostics:



Model diagnostics evaluate whether the key assumptions of linear regression—linearity, constant variance of residuals, normality, and absence of influential points—are satisfied. They help detect violations, outliers, or high-leverage observations, guiding model adjustments to ensure reliable inference and predictions.

i. **Residuals vs Fitted:**

Residuals show a funnel pattern (spread increases with fitted values), suggests heteroscedasticity variance of residuals increases for higher predicted incomes.Also some non-random curvatures indicate possible non-linear relationship not fully captured by the model.

ii. **Q-Q Residuals:**

Most points follow the line, but tails deviate, especially the upper tail. Residuals are slightly skewed, particularly for high incomes.

iii. **Scale-Location (Spread-Location):**

The red line slopes upward indicating variance increases with fitted values Confirms heteroscedasticity observed in Residuals vs Fitted.

iv. **Residuals vs Leverage:**

Most points are well within limits. A few points at high leverage or extreme residuals exist, but none appear extremely influential.

- **Multicollinearity Using VIF:**

```
                             GVIF Df GVIF^(1/(2*Df))
Age                      1.824808  1        1.350854
Total.Working.Years      1.982842  1        1.408134
Gender                   1.010724  1        1.005348
Years.Since.Last.Promotion 1.176505  1        1.084668
Department               1.011331  2        1.002821
Education                1.080702  4        1.009749
```

All predictors have GVIF^(1/(2*Df)) < 2, so there is no evidence of multicollinearity in the model.

Predictor estimates are reliable, and coefficients are not inflated due to correlation among predictors.

- **Autocorrelation (Durbin-Watson Test)**

H0 :ρ = 0 V.s H1:ρ > 0

```
	Durbin-Watson test

data:  lm_raw
DW = 2.0556, p-value = 0.8135
alternative hypothesis: true autocorrelation is greater than 0
```

Since p-value > 0.05, fail to reject H_0. Residuals are independent.

- **Linear model after log transformation**

The initial linear regression model of Monthly Income revealed violations of key assumptions, including heteroscedasticity and skewed residuals, as well as a highly variable scale of the response. To address these issues, a logarithmic transformation of Monthly Income was applied.

log_incomei =β0+β1(Agei)+β2(Total Working Yearsi)+β3(Genderi)+β4 (Years Since Last Promotioni)+β5 (DepartmentR&D_i)+ β6(DepartmentSales_i)+ β7(EducationBachelors_Degree_i)+β8(EducationDoctoral_Degree_i)+ β9 (Education. High_School_i) + β10(EducationMasters_Degree_i) + εi

The fitted model:

$(\widehat{\text{logMonthly Income}_i})$ = 7.8711 – 0.003186 (Agei) + 0.063749

(Total Working Yearsi) -0.007937 (GenderMalei)+ 0.008747(Years Since Last Promotioni) - 0.003033DepartmentR&Di) + 0.205948 (DepartmentSales)i + 0.010459(Education Bachelor_i)+ 0.145637(Education. Doctoral_i)+ 0.002087

(Education. High_School_i) +0.018468 (Education Master_i) .

```
Call:
lm(formula = log_income ~ Age + Total.Working.Years + Gender + 
    Years.Since.Last.Promotion + Department + edureg, data = train_data)

Residuals:
     Min       1Q   Median       3Q      Max 
-1.52353 -0.31581  0.02994  0.30514  1.31945 

Coefficients:
                            Estimate Std. Error t value Pr(>|t|)    
(Intercept)                 7.871108   0.102338  76.913  < 2e-16 ***
Age                        -0.003186   0.002096  -1.520  0.12879    
Total.Working.Years         0.063749   0.002623  24.306  < 2e-16 ***
GenderMale                 -0.007937   0.028823  -0.275  0.78309    
Years.Since.Last.Promotion  0.008747   0.004829   1.811  0.07037 .  
DepartmentR&D              -0.003033   0.073213  -0.041  0.96697    
DepartmentSales             0.205948   0.075786   2.718  0.00669 ** 
eduregBachelor's Degree     0.010459   0.038218   0.274  0.78439    
eduregDoctoral Degree       0.145637   0.080302   1.814  0.07003 .  
eduregHigh School           0.002087   0.052359   0.040  0.96822    
eduregMaster's Degree       0.018468   0.040952   0.451  0.65210    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4491 on 1018 degrees of freedom
Multiple R-squared:  0.5378,	Adjusted R-squared:  0.5332 
F-statistic: 118.4 on 10 and 1018 DF,  p-value: < 2.2e-16
```

- **Testing The Whole Model Significance Using F-test :**

H0: $\beta_1=\beta_2=\ldots\ldots= \beta_{10} = 0$ V.s H1:At least one $\beta_j \neq 0$

Since F on 10 df = 118.4, $p < 0.05$, indicates strong evidence against the null hypothesis that all regression coefficients are zero. The overall model is statistically significant. At least one predictor contributes meaningfully to explaining log income variation.

Adjusted R^2 = 0.5332 (~53.3%)

About 53% of the variation in log income is explained by the model predictors.

Residuals

The residuals are approximately symmetric and centered near zero, with a reduced spread compared to the raw model. This indicates that the log transformation stabilized variance, reduced skewness, and improved adherence to linear regression assumptions.

Coefficients' Interpretation:

- **Intercept ($\hat{\beta}_0$ = 7.8711, $p < 0.001$)**

Statistically significant. Represents the expected **log-income** for a **female employee in HR with an Associate Degree**, Age = 0, Total Working Years = 0, and Years Since Last Promotion = 0. This serves as the **baseline reference point**.

- **Total Working Years ($\hat{\beta}_2$ = 0.06375, $p < 0.001$)**

Statistically significant. Each additional year of total work experience increases **log-income** by approximately **6.4%**, holding other variables constant

- **DepartmentSales ($\hat{\beta}_6$ = 0.20595, $p = 0.0067$)**

Statistically significant. Employees in **Sales** earn approximately **22.8% higher income** than employees in HR (reference category), controlling for other predictors

Marginally significant predictor

- **Doctoral Degree ($\hat{\beta}$ = 0.14564, $p \approx 0.070$)**

Indicates that employees with a **doctoral degree** tend to earn higher income relative to the reference education level (Associate Degree), though evidence is weak (~15.7% increase).

- **Years Since Last Promotion ($\hat{\beta}_4$ = 0.00875, $p \approx 0.070$)**

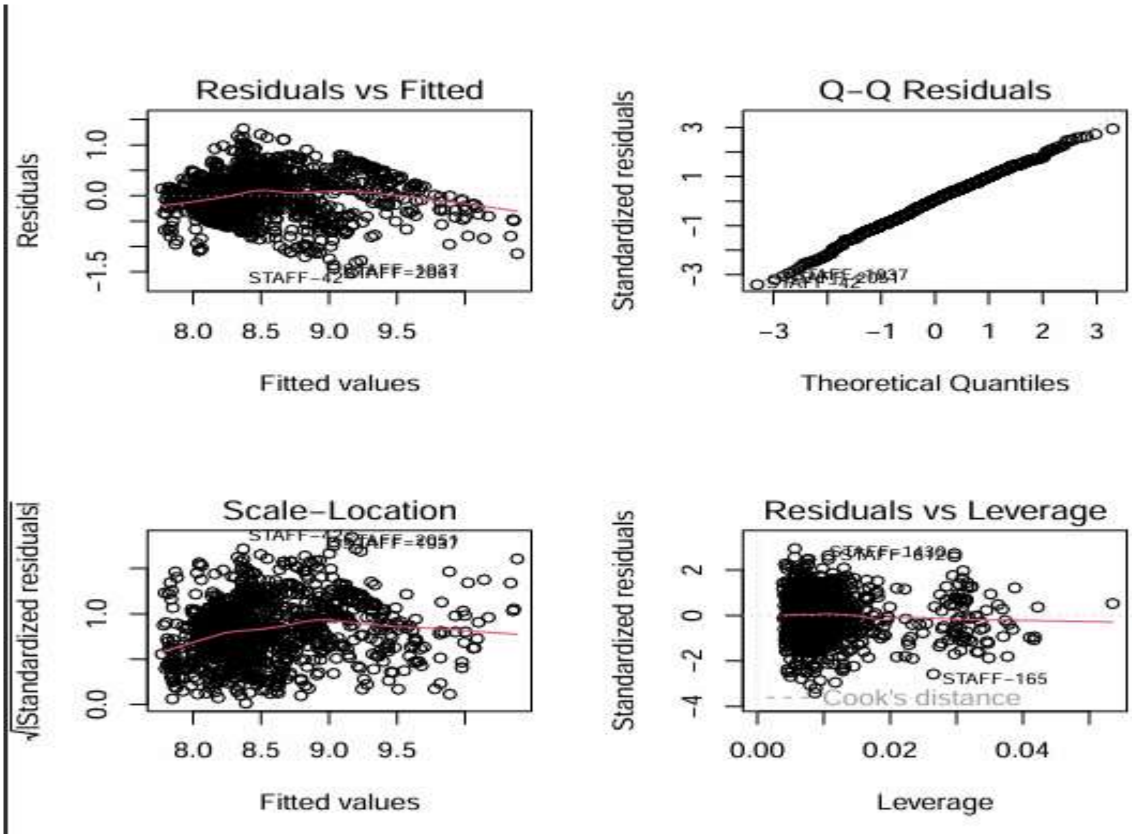
Suggests a small positive effect on log-income (~0.9% per year), but marginal significance indicates **weak evidence**.

Non-significant predictors ($\alpha = 0.05$)

Age ($\hat{\beta}_1 = -0.00319$), GenderMale ($\hat{\beta}_3 = -0.00794$), DepartmentR&D ($\hat{\beta}_5 = -0.00303$), Bachelor's Degree ($\hat{\beta} = 0.01046$), High School ($\hat{\beta} = 0.00209$), Master's Degree ($\hat{\beta} = 0.01847$)

These variables are **not statistically significant**, indicating **no reliable effect on log-income** once other factors are controlled.

Model Diagnostic:



i. **Residuals vs Fitted:**

Residuals are fairly randomly scattered around 0, with no clear pattern.Variance of residuals appears more constant than in the raw income model. Log transformation improved linearity and stabilized variance.

ii. **Q-Q Residuals:**

Most points lie along the 45° line. Slight deviations at the tails, but overall distribution is close to normal. Residuals are reasonably normal, supporting valid inference.

iii. **Scale-Location (Spread-Location)**

The red trend line is nearly horizontal. Residual spread is fairly constant across fitted values. Log transformation successfully reduced heteroscedasticity.

iv. **Residuals vs Leverage:**

Most points are within normal bounds. No extreme high-leverage points dominate the model. No single observation influences the log-linear model.

➢ **Multicollinearity Using VIF:**

The **GVIF^(1/(2*Df))** values for all predictors range from ~1.005 to 1.41. All values are well below the common threshold of 5, indicating **no evidence of problematic multicollinearity**. The predictors in the log-transformed model are not highly correlated, so the coefficient estimates are stable and reliable.

```
                            GVIF Df GVIF^(1/(2*Df))
Age                     1.824808  1        1.350854
Total.Working.Years     1.982842  1        1.408134
Gender                  1.010724  1        1.005348
Years.Since.Last.Promotion 1.176505  1        1.084668
Department              1.011331  2        1.002821
Education               1.080702  4        1.009749
```

➢ **Autocorrelation (Durbin-Watson Test)**

H0 :ρ = 0 V.s H1:ρ > 0

```
        Durbin-Watson test

data:  lm_log
DW = 2.059, p-value = 0.828
alternative hypothesis: true autocorrelation is greater than 0
```

The high p-value (> 0.05) confirms that we **fail to reject the null hypothesis** of no positive autocorrelation.

- **Stepwise Regression Using AIC**

Stepwise regression is a variable selection method that iteratively adds or removes predictors to identify a **parsimonious model** that balances **goodness-of-fit** and **model complexity**. In this analysis, we applied **both forward and backward stepwise selection** based on the **Akaike Information Criterion (AIC)**, which penalizes unnecessary variables to avoid overfitting. The process evaluates each candidate predictor's contribution and selects the combination of variables that **minimizes AIC**.

log(Monthly Incomei)= β0+ β1.Agei+ β2·Total Working Yearsi+ β3 Years Since Last Promotioni+β4DepartmentR&Di + β5.DepartmentSalesi+ εi

<u>The fitted model:</u>

$$\log(\widehat{\text{Monthly Income}_i}) = 7.8673 - 0.00301\,\text{Age}_i + 0.0640\,\text{Total Working Years}_i + 0.00879\,\text{Years Since Last Promotion}_i + 0.000327\,\text{DepartmentR\&D}_i + 0.2108102\,\text{DepartmentSales}_i$$

```
Call:
lm(formula = log_income ~ Age + Total.Working.Years + Years.Since.Last.Promotion +
    Department, data = train_data)

Residuals:
     Min       1Q   Median       3Q      Max 
-1.53412 -0.31235  0.02594  0.30872  1.34709 

Coefficients:
                             Estimate Std. Error t value Pr(>|t|)    
(Intercept)                 7.8673486  0.0949941  82.819  < 2e-16 ***
Age                        -0.0030128  0.0020520  -1.468  0.14235    
Total.Working.Years         0.0640247  0.0026130  24.503  < 2e-16 ***
Years.Since.Last.Promotion  0.0087894  0.0048248   1.822  0.06879 .  
DepartmentR&D               0.0003274  0.0730490   0.004  0.99642    
DepartmentSales             0.2108102  0.0755274   2.791  0.00535 ** 
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.4488 on 1023 degrees of freedom
Multiple R-squared:  0.5361,	Adjusted R-squared:  0.5339 
F-statistic: 236.5 on 5 and 1023 DF,  p-value: < 2.2e-16
```

- **Testing The Whole Model Significance Using F-test :**

H0: $\beta1=\beta2=\ldots\ldots.= \beta5= 0$ V.s H1:At least one $\beta j \neq 0$

Since F on 5 df = 236.5, $p < 0.05$, indicates strong evidence against the null hypothesis that all regression coefficients are zero. The overall model is statistically significant. At least one predictor contributes meaningfully to explaining log income variation.

Adjusted R^2 = 0.5339 (~53.4%)

Approximately 53% of the variability in log income is explained by the predictors in this stepwise-selected model.

Residuals:

Residuals are approximately symmetrically distributed around zero, indicating the model is unbiased on average. Spread is moderate and uniform, suggesting the log transformation has successfully stabilized variance. No extreme outliers dominate the model, supporting assumption validity.

Coefficients' Interpretation:

- **Intercept (β_0 = 7.8673, $p < 0.001$)**

Statistically significant. Represents the expected **log-income** for a **female employee in HR with an Associate Degree**, Age = 0, Total Working Years = 0, and Years Since Last Promotion = 0. Serves as the **baseline reference point**.

- **Total Working Years (β_2 = 0.06402, $p < 0.001$)**

Statistically significant. Each additional year of work experience increases log-income by **6.6%** ($e^{0.06402} - 1$), holding other variables constant. This remains the **most influential predictor**.

- **DepartmentSales ($\hat{\beta}_5$ = 0.21081, p = 0.00535)**

Statistically significant. Employees in **Sales** earn ~23.4% higher income than those in HR (reference category), controlling for experience, age, and promotion history

Marginally significant predictor

- **Years Since Last Promotion ($\hat{\beta}_3$ = 0.00879, p ≈ 0.069)**

Slight positive effect (~0.9% per year), indicating weak evidence that time since last promotion may slightly increase income.

Non-significant predictors

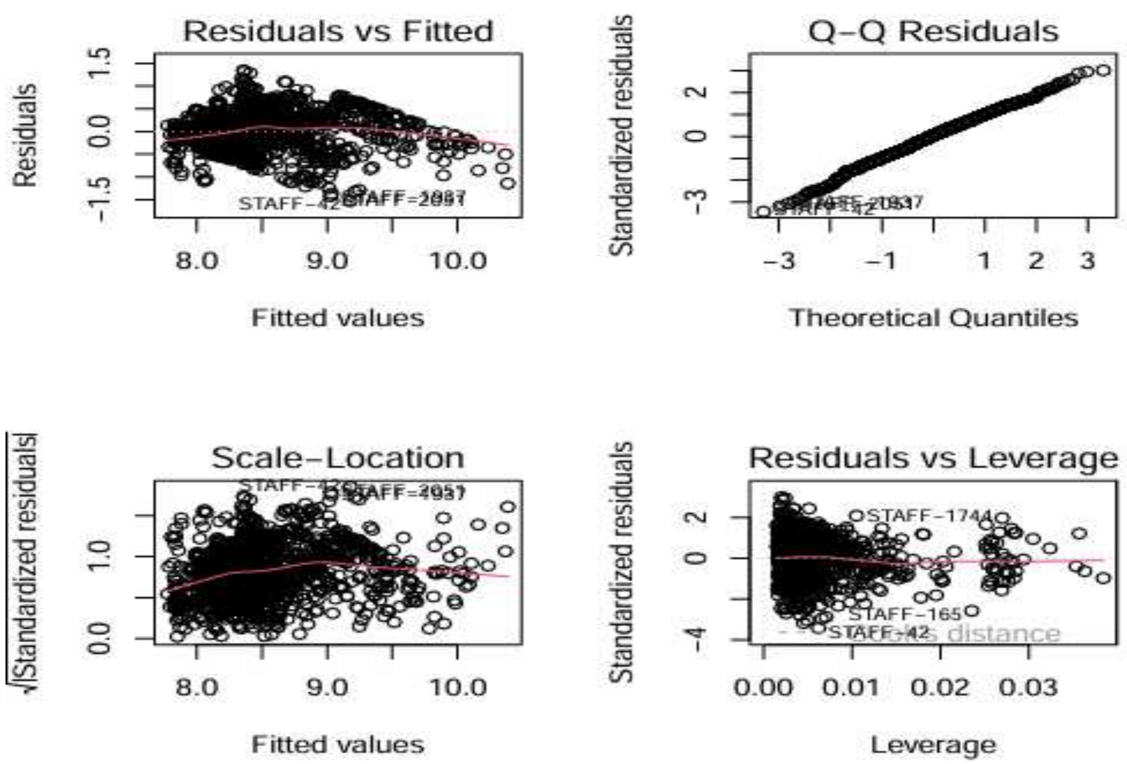
- **DepartmentR&D ($\hat{\beta}_4$ = 0.00033, p = 0.996)**

Employees in R&D do not earn significantly different log-income compared to HR employees.

- **Age ($\hat{\beta}_1$ = −0.00301, p = 0.142)**

Age has no reliable effect on log-income after accounting for experience, promotion, and department.

Model Diagnostics:



i. Residuals vs Fitted:

Residuals are scattered fairly randomly around 0, without a clear pattern. Variance appears roughly constant across fitted values. The model shows **linear relationships** and **homoscedasticity** is reasonably satisfied after log transformation.

ii. Q-Q Residuals:

Most points lie along the 45° line.Slight deviations at the tails, but overall residuals are close to normal.Normality assumption is reasonably satisfied, supporting valid t-tests and p-values.

iii. **Scale-Location (Spread-Location)**

The red trend line is nearly horizontal. Spread of residuals is roughly constant across fitted values. Confirms that variance of residuals is stabilized in the stepwise model.

iv. Residuals vs Leverage:

Most points are within normal bounds; no extreme leverage points dominate the model. No single observation is overly influential; model is robust.

- **Multicollinearity Using VIF :**

All values are well below the common threshold of 5, indicating **no evidence of problematic multicollinearity**.

```
                             GVIF Df GVIF^(1/(2*Df))
Age                          1.751205  1        1.323331
Total.Working.Years          1.970737  1        1.403829
Years.Since.Last.Promotion 1.176127  1        1.084494
Department                   1.003757  2        1.000938
```

- **Autocorrelation (Durbin-Watson Test)**

H0 :ρ = 0 V.s H1:ρ > 0

```
        Durbin-Watson test

data:  lm_step
DW = 2.0523, p-value = 0.7997
alternative hypothesis: true autocorrelation is greater than 0
```

Since p-value > 0.05, fail to reject H0. Residuals are independent, satisfying the linear regression assumption of uncorrelated errors.

Model Evaluation on Test Data

After fitting the final stepwise-selected regression model on the cleaned training data, it is important to assess its **predictive performance** on unseen data. Model evaluation on a **hold-out test set** provides an unbiased estimate of how well the model generalizes to new observations. Common evaluation metrics for regression models include the **Root Mean Squared Error (RMSE)**, which measures the average magnitude of prediction errors, and the **R-squared (R^2)**, which indicates the proportion of variance in the response variable explained by the model.

- **Root Mean Squared Error (RMSE):** 0.406

```
> rmse_lm
[1] 0.4060863
```

This value represents the average difference between the predicted and actual log-income values in the test set. A smaller RMSE indicates more accurate predictions; in this case, an RMSE of 0.406 suggests that, on average, the model's predictions deviate from the true log-income by roughly 0.41 units.

- **R-squared (R^2):** 0.639 (~63.9%)

```
> r2_lm
[1] 0.6439993
```

meaning that the model explains roughly **64% of the variation in log-income** among employees not used to fit the model. This demonstrates that the model generalizes fairly well to unseen data, capturing the main patterns while leaving some residual variability that may be due to unobserved factors such as individual performance, bonuses, or other personal characteristics.

Results:

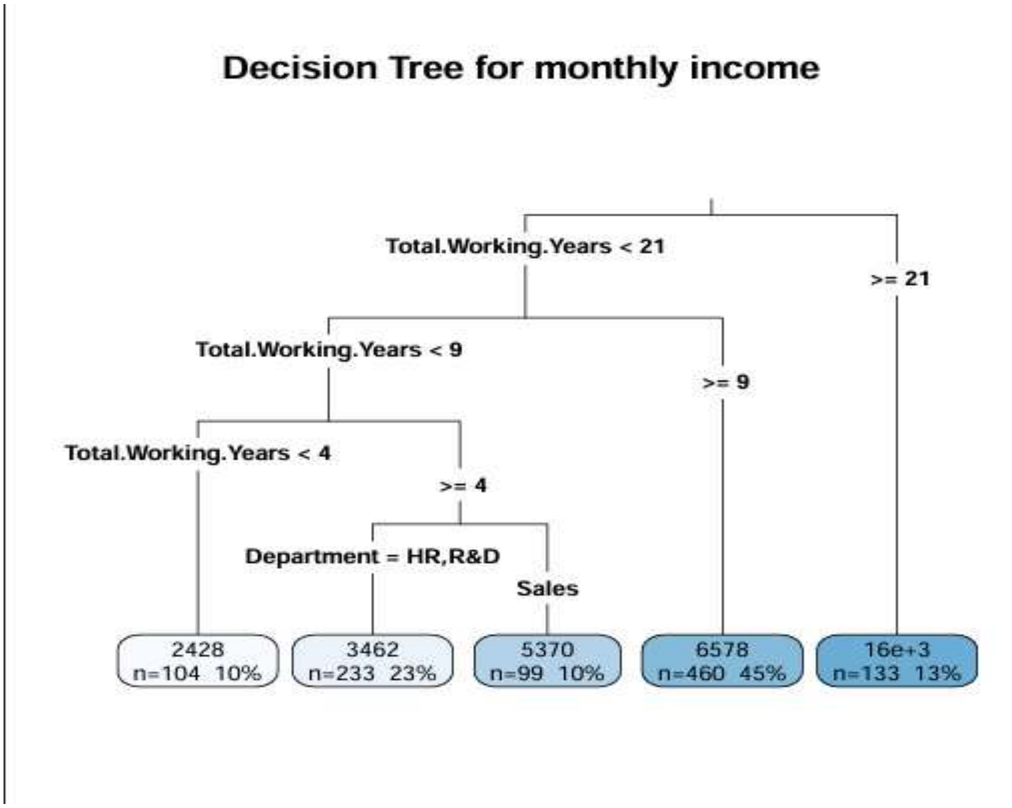
The regression analysis demonstrates that employee monthly income can be reasonably explained by a set of demographics, professional, and organizational variables, with clear evidence that work experience plays the most dominant role. The initial linear regression model revealed violations of key assumptions, particularly heteroscedasticity and skewed residuals, which justified the use of a logarithmic transformation of the response variable.

After applying the log transformation, the model exhibited substantial improvement in model diagnostics, including more stable variance, approximately normal residuals, and reduced influence of extreme observations. The transformed model confirmed that Total Working Years is the most statistically significant predictor of income, indicating that each additional year of experience leads to a meaningful percentage increase in earnings, holding all other factors constant.

the results suggest that departmental affiliation matters in specific cases, as employees working in the Sales department tend to earn higher incomes compared to those in Human Resources, even after controlling for experience and other characteristics. In contrast, variables such as gender, age, and most education levels did not show statistically significant effects once experience was accounted for, implying that income determination in this organization is primarily experience-driven rather than demographically driven.

ML Algorithm (Decision Tree):

In this analysis, a **decision tree regression model** was trained to predict **Monthly Income** using employee characteristics, including Age, Total Working Years, Years Since Last Promotion, Gender, Department, and Education. The model recursively splits the training data based on predictor values to minimize the variance within each node, creating a hierarchical structure that assigns predicted income values to the terminal nodes.



Root node (most important split)

Total.Working.Years

This is the **primary determinant** of income. Employees are first separated into:

- **Less experienced (< 21 years)**
- **Highly experienced (≥ 21 years)**

Right branch: Highly experienced employees

Total.Working.Years ≥ 21

- **Predicted income ≈ 16000**
- **13% of employees (n = 133)**

Employees with **21 or more years of experience earn the highest income**, regardless of department. Experience dominates all other factors here.

Left branch: Less experienced employees (< 21 years)

i. **Total.Working.Years < 9**

a) Total.Working.Years < 4

- **Predicted income** = 2428
- **10% of employees (n = 104)**

Very early-career employees earn the **lowest income**, with department having no effect

b) 4 ≤ Total.Working.Years < 9

This group is further split by **Department**.

- **HR & R&D**
  - **Predicted income ≈ 3462**
  - **23% of employees (n = 233)**
- **Sales**
  - **Predicted income ≈ 5370**
  - **10% of employees (n = 99)**

For employees with **moderate early experience**, **department matters**:

- Sales employees earn **substantially more** than HR and R&D.
- This is the **only point** where department significantly influences income.

ii. **9 ≤ Total.Working.Years < 21**

- **Predicted income** ≈ 6578
- **45% of employees (n = 460)**

Mid-to-senior employees earn **high and stable income**, and department no longer plays a role.

The decision tree analysis confirms that **Total Working Years is the dominant predictor of monthly income**. Employees with **more experience earn higher incomes**, with those having ≥ 21 years reaching the top income bracket, regardless of department. For early-career employees (< 9 years), **department plays a role**, with Sales employees earning more than those in HR or R&D. Beyond 9 years of experience, income becomes more stable, and **department no longer**

significantly affects earnings. **Overall,** the tree highlights that **work experience drives income growth**, while department influences are limited to employees with moderate early experience.

<u>**Model Evaluation:**</u>

On the test dataset, the decision tree's predictive accuracy was evaluated using RMSE and R^2. The **RMSE** measures the average prediction error in the income units, indicating the typical deviation between predicted and observed income. The R^2value represents the proportion of variance in monthly income explained by the tree.

RMSE: The **RMSE** of the decision tree on the test data is **2,277.56**, indicating that, on average, the predicted monthly income deviates from the actual income by approximately 2,278 units. This reflects reasonably accurate predictions, considering the natural variability in employee income.

```
> rmse_tree
[1] 2277.557
```

$\boldsymbol{R^2}$**:** The model's $R^2 = 0.780587$, meaning that the model explains about **78% of the variance** in monthly income in the test set. This demonstrates that the decision tree captures the majority of the variability in income and provides strong predictive performance for unseen data.

```
> r2_tree
[1] 0.7805876
```

Main Results and Conclusion:

This project aimed to analyze and predict **employee monthly income** by integrating **classical statistical techniques with machine learning methods**, using a comprehensive HR dataset. Through descriptive analysis and exploratory data visualization, the study revealed a workforce characterized by a right-skewed income distribution, a mid-career majority, and a strong concentration of employees within the Research and Development department.

The regression analysis demonstrated that **total work experience is the primary determinant of income**, while most demographic variables have limited explanatory power once experience is controlled for. Model refinements, including logarithmic transformation and stepwise variable selection, improved both statistical validity and predictive performance, with the final regression model showing good generalization to unseen data.

Complementing the regression results, the **decision tree model provided a clear and intuitive representation of the income structure**, highlighting important non-linear relationships that are difficult to capture using traditional linear models. The tree confirmed that **Total Working Years is the dominant splitting variable**, separating employees into distinct income tiers. It also revealed that **departmental effects are mainly relevant during early career stages**, particularly benefiting employees in Sales, while income becomes more stable and less department-dependent as experience increases.

In summary, the combined use of regression and decision tree models provided a comprehensive assessment of how demographic, educational, and professional factors collectively influence employee monthly income. While these variables contribute to income prediction, the results consistently demonstrate that **total work experience is the most influential driver of compensation**, with demographic and departmental factors playing a secondary role, particularly during early career stages. This integrated analytical approach enhances interpretability, predictive accuracy, and practical relevance, offering robust evidence to support data-driven compensation and human resource policy decisions.

Declaration of Equal Contribution:

We hereby declare that **all four group members contributed approximately equally to this project**. The work was carried out collaboratively across all stages, including **data preparation, coding, statistical analysis, model implementation, interpretation of results, and report writing**. Each member actively participated in developing the regression models and the decision tree analysis, as well as in reviewing and refining the final report. Tasks were discussed and distributed collectively, with continuous coordination and mutual feedback throughout the project. Therefore, the final output represents a **joint effort**, and all group members share equal responsibility for the content and conclusions of this study.

APPENDIX:

```r
#####################################################
#      FEPS Data Analysis Project
# - Salma Khaled Ahmed Taiaa       (ID: 5220475)
# - Mona Ahmed Abdelmonem          (ID: 5220205)
# - Menna Tallah Mostafa ElSayed   (ID: 5220819)
# - Reham Mohamed Eldemerdash      (ID: 5220671)
#####################################################

# -------------------------------------------------------------
# Load required packages for data analysis and visualization
# -------------------------------------------------------------
library(psych)
library(DescTools)
library(corrplot)
library(vcd)
library(rpart)
library(rpart.plot)
library(car)
library(lmtest)

# -------------------------------------------------------------
# a) Read the HR dataset and select relevant variables
# -------------------------------------------------------------
df <- read.csv("HR Data.csv")
df <- df[, c(
  "emp.no",
  "Age",
  "Monthly.Income",
  "Gender",
  "Department",
  "Total.Working.Years",
  "Years.Since.Last.Promotion",
  "Education"
)]

# -------------------------------------------------------------
# b) Explore dataset structure and basic summaries
# -------------------------------------------------------------
head(df)
names(df)
str(df)
summary(df)
dim(df)
```

```r
# ------------------------------------------------------------
# Convert categorical variables to factor type
# ------------------------------------------------------------
df$edureg <- as.factor(df$Education) #so the variable will appear with it's categories in the reggression model
df$Education <- factor(
  df$Education,
  levels = c(
    "High School",
    "Associates Degree",
    "Bachelor's Degree",
    "Master's Degree",
    "Doctoral Degree"
  ),
  ordered = TRUE
)
df$Department <- as.factor(df$Department)
df$Gender <- as.factor(df$Gender)

# ------------------------------------------------------------
# Assign employee number as row names and remove ID column
# ------------------------------------------------------------
rownames(df) <- df$emp.no
df$emp.no <- NULL

# ------------------------------------------------------------
# Re-check data structure after transformations
# ------------------------------------------------------------
dim(df)
summary(df)

# ------------------------------------------------------------
# Perform descriptive statistical analysis
# ------------------------------------------------------------
Desc(df)

# ------------------------------------------------------------
# Custom function for categorical variables:
# Computes mode and relative frequency
# ------------------------------------------------------------
summary_cat <- function(data) {

  results <- list()

  for (var in names(data)) {
    x <- data[[var]]

    if (is.factor(x) || is.character(x)) {
      tab <- table(x)
      results[[var]] <- list(
        type = "Qualitative",
        mode = names(tab)[which.max(tab)],
```

```r
      results[[var]] <- list(
        type = "Qualitative",
        mode = names(tab)[which.max(tab)],
        relative_frequency = prop.table(tab)
      )
    }
  }

  return(results)
}

summary_cat(df)

# ------------------------------------------------------------
# Correlation analysis for numeric variables
# ------------------------------------------------------------
numeric_vars <- df[, sapply(df, is.numeric)]
cor_matrix <- cor(numeric_vars)

corrplot(
  cor_matrix,
  method = "color",
  type = "upper",
  diag = FALSE,
  addCoef.col = "black",
  number.cex = 0.6,
  tl.col = "black",
  tl.cex = 0.8,
  tl.srt = 45,
  na.label = " "
)

# ------------------------------------------------------------
# Mosaic plot: Association between Department, Education, and Gender
# ------------------------------------------------------------
tab1 <- table(
  Department = df$Department,
  Education  = df$Education,
  Gender     = df$Gender
)

mosaic(
  tab1,
  shade = TRUE,
  legend = TRUE,
  labeling = labeling_border,
  suppress = TRUE,
  rot_labels = c(0, 0, 0, 90),
  main = "Association of Department, Education, and Gender"
)
```

```r

# ----------------------------------------------------------
# Custom Function: Automatic Exploratory Data Visualization
# ----------------------------------------------------------
auto_plots <- function(data) {

  for (var in names(data)) {

    x <- data[[var]]

    if (is.numeric(x)) {

      par(mfrow = c(1, 2),
          mar = c(4, 4, 3, 1),
          bg = "gray95")

      hist(
        x,
        main = paste("Histogram of", var),
        xlab = var,
        col = "plum",
        border = "gray40",
        breaks = 12
      )

      boxplot(
        x,
        main = paste("Boxplot of", var),
        col = "pink",
        horizontal = TRUE,
        border = "gray40"
      )

      par(mfrow = c(1, 1))

    } else if (is.factor(x) || is.character(x)) {

      freq <- table(x)

      par(mfrow = c(1, 2),
          mar = c(7, 4, 3, 1),
          bg = "gray95")

      bp <- barplot(
        freq,
        main = paste("Bar Chart of", var),
        col = "gray70",
        border = "gray30",
        las = 2,
        ylim = c(0, max(freq) * 1.2)
      )
```

```r

      text(
        x = bp,
        y = freq,
        label = freq,
        pos = 3,
        cex = 0.8,
        col = "purple4"
      )

      pie(
        freq,
        main = paste("Pie Chart of", var),
        col = c("pink", "plum", "thistle", "gray70", "lavender")
      )

      par(mfrow = c(1, 1))
    }
  }
}

# ----------------------------------------------------------------
# Apply automatic plotting function to the dataset
# ----------------------------------------------------------------
auto_plots(df)

# ----------------------------------------------------------------
# DATA CLEANING
# ----------------------------------------------------------------

# Check for missing values
colSums(is.na(df))

# Visual inspection of outliers using boxplots
par(mfrow = c(4, 1))
for (i in names(numeric_vars)) {
  boxplot(
    numeric_vars[[i]],
    main = i,
    col = "lightgray",
    horizontal = TRUE
  )
}
par(mfrow = c(1, 1))
```

```r

######################################################
#        Linear Regression (Train/Test Framework)
######################################################
# ------------------------------------------------------------
# Log transformation of income
# ------------------------------------------------------------
df$log_income <- log(df$Monthly.Income) # as it's money so it's unit is very high

# ------------------------------------------------------------
# Train / Test Split (70% / 30%)
# ------------------------------------------------------------
set.seed(789)

train_index <- sample(
  seq_len(nrow(df)),
  size = 0.7 * nrow(df)
)

train_data <- df[train_index, ]
test_data  <- df[-train_index, ]


# ------------------------------------------------------------
# Initial Linear Model (Raw-Training Data)
# ------------------------------------------------------------
lm_raw <- lm(
  Monthly.Income ~ Age + Total.Working.Years + Gender +
    Years.Since.Last.Promotion + Department + edureg,
  data = train_data
)

summary(lm_raw)

par(mfrow = c(2, 2))
plot(lm_raw)
par(mfrow = c(1, 1))
# Multicollinearity check
# ------------------------------------------------------------
vif(lm_raw)
dwtest(lm_raw)
# ------------------------------------------------------------
# Linear model after log transformation
# ------------------------------------------------------------
lm_log <- lm(
  log_income ~ Age + Total.Working.Years + Gender +
    Years.Since.Last.Promotion + Department + edureg,
  data = train_data
)

summary(lm_log)

```

```r
par(mfrow = c(2, 2))
plot(lm_log)
par(mfrow = c(1, 1))

# ------------------------------------------------------------
# Multicollinearity check
# ------------------------------------------------------------
vif(lm_log)
dwtest(lm_log)
# ------------------------------------------------------------
# Stepwise regression using AIC
# ------------------------------------------------------------
lm_step <- step(lm_log, direction = "both", trace = FALSE)
summary(lm_step)

# ------------------------------------------------------------
# Diagnostic plots for Stepwise model
# ------------------------------------------------------------
par(mfrow = c(2, 2))
plot(lm_step)
par(mfrow = c(1, 1))

# ------------------------------------------------------------
# Multicollinearity - Stepwise model
# ------------------------------------------------------------
vif(lm_step)

# ------------------------------------------------------------
# Independence of errors
# ------------------------------------------------------------
dwtest(lm_step)



#####################################################
#        Model Evaluation (Test Data)
#####################################################

lm_pred <- predict(lm_step, newdata = test_data)

rmse_lm <- sqrt(mean((lm_pred - test_data$log_income)^2))

sst <- sum((test_data$log_income - mean(test_data$log_income))^2)
sse <- sum((test_data$log_income - lm_pred)^2)
r2_lm <- 1 - sse / sst

rmse_lm
r2_lm


#####################################################
```

```r
##########################################################
#          Machine Learning Preparation
##########################################################

ml_train <- train_data[, c(
  "Monthly.Income",
  "Age",
  "Total.Working.Years",
  "Years.Since.Last.Promotion",
  "Gender",
  "Department",
  "Education"
)]

ml_test <- test_data[, c(
  "Monthly.Income",
  "Age",
  "Total.Working.Years",
  "Years.Since.Last.Promotion",
  "Gender",
  "Department",
  "Education"
)]

##########################################################
#          Decision Tree Regression Model
##########################################################

tree_model <- rpart(
  Monthly.Income ~ .,
  data = ml_train,
  method = "anova"
)

summary(tree_model)

##########################################################
#          Visualize the decision tree
##########################################################

rpart.plot(
  tree_model,
  type = 3,
  extra = 101,
  fallen.leaves = TRUE,
  main = "Decision Tree for Monthly.Income"
)

#######################################################
#          Predict & Evaluate (Test Data)
#######################################################

tree_pred <- predict(tree_model, newdata = ml_test)

rmse_tree <- sqrt(mean((tree_pred - ml_test$Monthly.Income)^2))

sst <- sum((ml_test$Monthly.Income - mean(ml_test$Monthly.Income))^2)
sse <- sum((ml_test$Monthly.Income - tree_pred)^2)
r2_tree <- 1 - sse / sst

rmse_tree
r2_tree
```